

Cool Maths

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1 Introduction

This document provides (some) of my solutions to several problems, mostly from the Putnam and IMO competitions. Several other solutions are available, and will be slowly added in. They were written with the purpose to serve as an example for others to understand my own thought process, and originally (and still) were shared through Discord, which is the reason they are mostly in an image format.

2 Putnam 1985 B-3

Problem Let

$$\begin{array}{cccc} a_{1,1} & a_{1,2} & a_{1,3} & \cdots \\ a_{2,1} & a_{2,2} & a_{2,3} & \cdots \\ a_{3,1} & a_{3,2} & a_{3,3} & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{array}$$

be a doubly infinite matrix of positive integers such that every positive integer appears exactly 8 times in it. Show that $\exists a_{m,n}$ such that $a_{m,n} > mn$.

Proof. We show a contradiction by size. Suppose the matrix indeed verified that no entry $a_{m,n}$ is greater than mn . Let k be a positive integer and consider the number 2^k . This number can be factorised into pairs of terms of the form $(2^x, 2^{k-x})$, for $x \in \{0, 1, \dots, k\}$. Now, it is clear that, given $a_{m,n}$ in the matrix, every number in the rectangle:

$$\begin{array}{ccc} a_{1,1} & \cdots & a_{1,n} \\ \vdots & \ddots & \vdots \\ a_{m,1} & \cdots & a_{m,n} \end{array} \quad \begin{array}{c} \cdots \\ \ddots \end{array}$$

Must be at most mn , as its product of indices is at most exactly mn . Now, we consider, for each k , the area represented by the union of all of these rectangles for each factorisation of 2^k :

$a_{1,1}$	$a_{1,2}$	$a_{1,3}$	$a_{1,4}$	$\cdots$	$a_{1,2^{k-2}}$	$\cdots$	$a_{1,2^{k-1}}$	$\cdots$	$a_{1,2^k}$
$a_{2,1}$	$\ddots$				$\vdots$		$a_{2,2^{k-1}}$		
$a_{3,1}$		$\ddots$			$\vdots$		$\vdots$		
$a_{4,1}$			$\ddots$		$a_{4,2^{k-2}}$				
$\vdots$					$\vdots$				
$a_{2^{k-2},1}$	$\cdots$	$\cdots$	$a_{2^{k-2},4}$	$\cdots$					
$\vdots$									
$a_{2^{k-1},1}$	$a_{2^{k-1},2}$	$\cdots$							
$\vdots$									
$a_{2^k,1}$									

The result of the total area will be denoted as $A(2^k)$. The computation of it directly is irrelevant to us at this point in time; we will fixate ourselves in the *step* at each consecutive power, that is, we want to compute

$$A(2^k) - A(2^{k-1})$$

This is the lightest grey area, and is the sum of all rectangles "added" at each step; we denote it by $S(2^k)$. To compute each individual light grey rectangle, we consider its vertices: namely, $a_{2^{x-1}, 2^{k-x-1}}, a_{2^x, 2^{k-x-1}}, a_{2^x, 2^{k-x}}, a_{2^{x-1}, 2^{k-x}}$ (for the sake of our eyes, we will denote them from here just like normal points in the plane, that is, (x, y)). There are degenerate cases when $x = 0, k$, as the rectangles virtually have two pairs of vertices coincide. Observant readers may notice only one of the vertices actually lies in the rectangle, and the rest lie in the slightly darker rectangles; but this precisely cancels the fact that the vertices technically occupy area (each entry in the matrix can be thought of as one unit of area); thus, the area can be computed directly:

$$\begin{aligned}
& \text{Area}((2^{x-1}, 2^{k-x-1}), (2^x, 2^{k-x-1}), (2^x, 2^{k-x}), (2^{x-1}, 2^{k-x})) = \\
& = \text{Base} \cdot \text{Height} = (2^x - 2^{x-1}) \cdot (2^{k-x} - 2^{k-x-1}) = 2^{x-1} \cdot 2^{k-x-1} = \\
& = 2^{k-2}
\end{aligned}$$

In total, the areas at the nondegenerate cases is always 2^{k-2} ; given there are $k-1$ such nondegenerate cases, the nondegenerate cases contribute $(k-1)2^{k-2}$ to the area.

The degenerate cases have bases $2^{k-1}, 1$ and heights $1, 2^{k-1}$, so they each contribute 2^{k-1} to the area. Thus, we obtain:

$$S(2^k) = (k-1)2^{k-2} + 2 \cdot 2^{k-1} = (k+3)2^{k-2}$$

We must remember that the numbers inside the total grey area are at most 2^k , so they must be from the set of numbers between 1 and 2^k . Given we have exactly 8 repetitions of each number, the numbers in the grey area can be any number from a pool of $8 \cdot 2^k$. Observe that this is the same as $32 \cdot 2^{k-2}$, which is smaller than *the step* from 2^{29} to 2^{30} ; thus, it is clearly smaller than $A(2^k)$ for some k , and there would be more numbers to fill than the number of *possible* numbers we can choose from, leading to a contradiction.

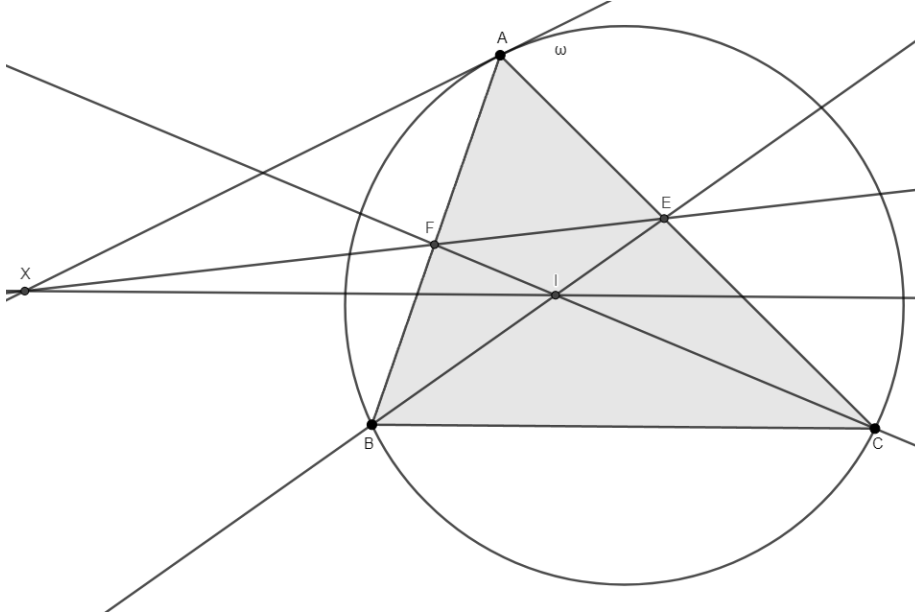
□

Side note. This problem chooses the number 8 arbitrarily; without almost any change to the proof (aka the last step), any constant c would be fine instead of 8. Also, as we tend the power of 2 to infinity, the area begins to look like the integral of $\frac{2^k}{x}$; in particular, this solution shows that the approximations of this area grow faster than a constant multiple of 2^k .

3 Cayley-Bacharach

The Cayley-Bacharach Theorem states that if two cubics intersect at eight points $P_1, P_2, P_3, P_4, P_5, P_6, P_7, P_8$ and at a point P then any other cubic through $P_1, P_2, P_3, P_4, P_5, P_6, P_7, P_8$ also goes through P .

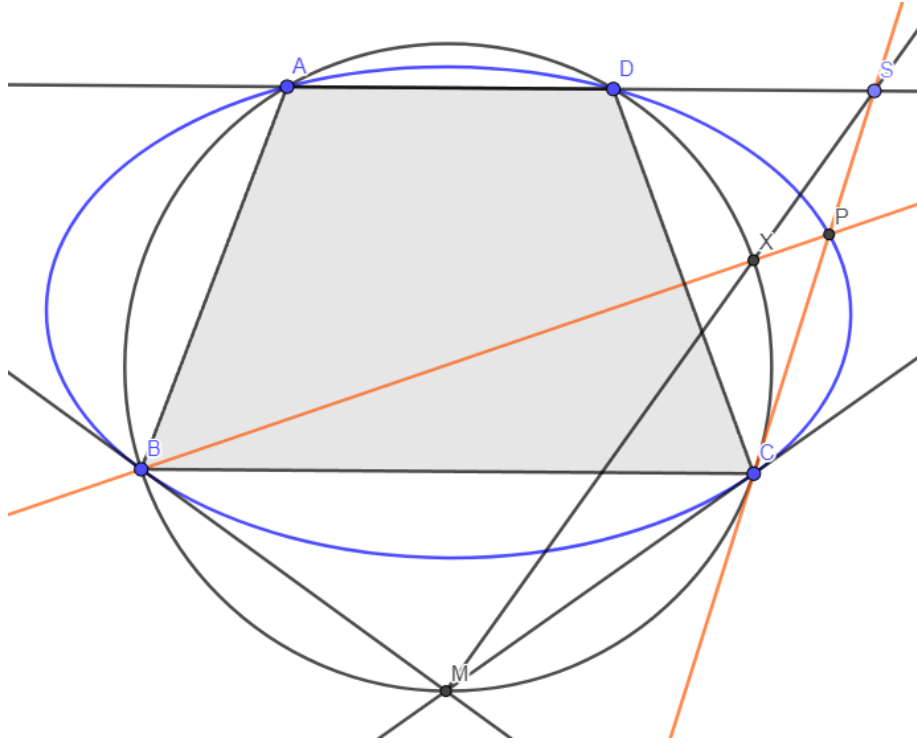
Problem Let $\triangle ABC$ be a triangle, Ω its circumcircle and I its incenter. Let r be the line through I parallel to $\overline{BC}$, and E, F the intersections of $\overline{BI} \cap \overline{AC}$ and $\overline{CI} \cap \overline{AB}$, respectively. Let $X = r \cap \overline{EF}$. Show $\overline{AX}$ is tangent to Ω .



Proof. Let S, T be the intersections of r with Ω . Consider the set of points $\{B, C, S, T, M_B, M_C, E, F\}$. The cubics $\overline{BM_B} \cup \overline{CM_C} \cup \overline{ST}$ (in red) and $\Omega \cup \overline{EF}$ (in green) both pass through all 8 points and intersect at X . By the Cayley-Bacharach Theorem, every cubic through our set of points passes through X . Our purpose now is to show that $EFSTBC$ is a conic (the ellipse in blue).

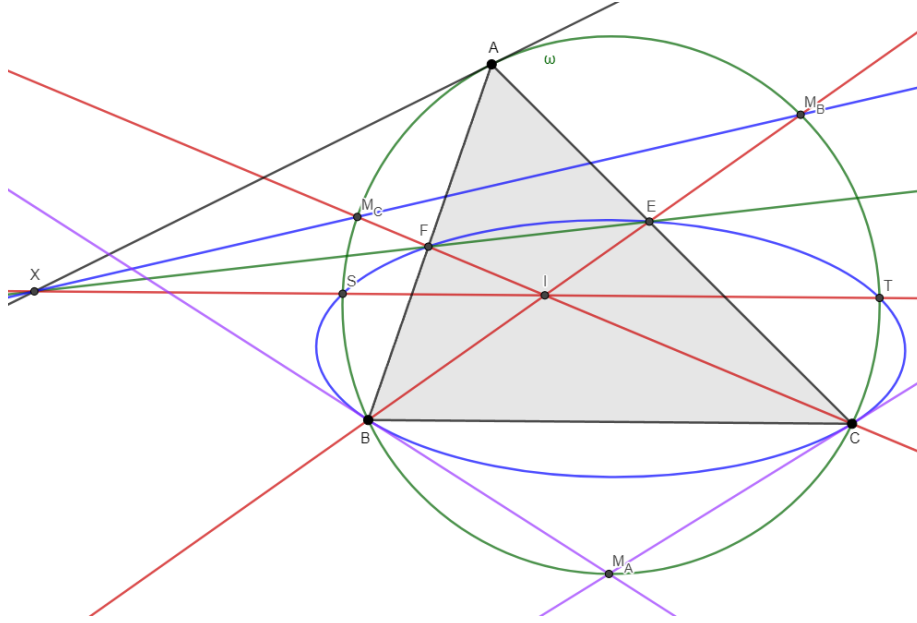


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Given this, we simply set $SBCT$ to be our quadrilateral and observe that $S = I$ is the case we are working with. By symmetry, we can see that E and F both lie in the ellipse.

Now, given $EFSTBC$ is a conic, we get the cubic $EFSTBC \cup \overline{M_B M_C}$ (all in blue). It is clear this cubic also includes the set of points $\{B, C, S, T, M_B, M_C, E, F\}$, and thus also goes through X ; in particular, $\overline{M_B M_C}$ goes through X (X cannot lie in the ellipse as it is defined as the intersection of two distinct lines on the conic that intersect it at four distinct points, unless the degenerate cases $E = S$ or $F = T$ are true).



It is now a routine task to prove the problem; consider the degenerate inscribed hexagon $AABM_BM_CC$ and Pascal's Theorem implies the tangent at A to Ω , the line $\overline{M_BM_C}$ and the line $\overline{EF}$ concur; but as we know, $\overline{M_BM_C} \cap \overline{EF} = X$, and we are done.

□

4 IMO SL N2 (Peru)

Problem. A positive integer N is called balanced, if $N = 1$ or if N can be written as a product of an even number of not necessarily distinct primes. Given positive integers a and b , consider the polynomial P defined by

$$P(x) = (x + a)(x + b)$$

1. Prove that there exist distinct positive integers a and b such that all the numbers $P(1), P(2), \dots, P(50)$ are balanced.
2. Prove that if $P(n)$ is balanced for all positive integers n , then $a = b$.

Proof. For the first statement, my first approach would be to prove there are sequences of unbalanced numbers of arbitrary length (or at least one of length 51). I have no clue how to further proceed though.

However, for the sake of completeness, I will feature the "intended" solution.

Define a function f on the set of positive integers by $f(n) = 0$ if n is balanced and $f(n) = 1$ otherwise. Clearly, $f(nm) \equiv f(n) + f(m) \pmod{2}$ for all positive integers n, m . Now for each positive integer n consider the binary sequence $(f(n+1), f(n+2), \dots, f(n+50))$. As there are only 2^{50} different such sequences there are two different positive integers a and b such that $(f(a+1), f(a+2), \dots, f(a+50)) = (f(b+1), f(b+2), \dots, f(b+50))$. But this implies that for the polynomial $P(x) = (x+a)(x+b)$ all the numbers $P(1), P(2), \dots, P(50)$ are balanced, since for all $1 \leq k \leq 50$ we have $f(P(k)) \equiv f(a+k) + f(b+k) \equiv 2f(a+k) \equiv 0 \pmod{2}$.

□

Side note. Very vaguely, what this solution does is quite similar to my approach to the second solution (with the use of a function), but using a combinatorial idea simply states that a finite sequence has finitely many possibilities, but as we have infinitely many numbers, at least one sequence must repeat. With the repeat, the polynomial becomes balanced.

(Cue in my own solution for the second part, which is similar but I believe more elementary)

For the second statement, we define a function $M_2(x)$ by the condition:

$$M_2(x) \begin{cases} 1 & \text{if } x \text{ is balanced} \\ 0 & \text{if } x = 0 \\ -1 & \text{otherwise} \end{cases}$$

It is clear that this function is multiplicative; that is:

$$M_2(xy) = M_2(x) \cdot M_2(y)$$

By definition, we thus obtain $M_2(P(x)) = 1$ for any x . Using this, we may observe a few things about a and b . Consider the cases $x = a$ and $x = b$:

$$\begin{aligned} M_2(P(a)) = 1 &\Rightarrow M_2((a+a)(a+b)) = M_2(2a) \cdot M_2(a+b) = \\ &= M_2(2) \cdot M_2(a) \cdot M_2(a+b) = 1 \Rightarrow M_2(a) \cdot M_2(a+b) = -1 \end{aligned}$$

$$M_2(P(b)) = 1 \Rightarrow M_2((a+b)(b+b)) = M_2(a+b) \cdot M_2(2b) =$$

$$= M_2(a+b) \cdot M_2(b) \cdot M_2(2) = 1 \Rightarrow M_2(a+b) \cdot M_2(b) = -1$$

Thus we have:

$$\begin{aligned} M_2(a) \cdot M_2(a+b) = -1 &= M_2(a+b) \cdot M_2(b) \Rightarrow M_2(a) = M_2(b) \Rightarrow \\ &\Rightarrow M_2(a)M_2(b) = M_2(b)^2 = M_2(b^2) = 1 \end{aligned}$$

So we conclude ab is balanced.

Now, we consider "translating" the pair (a, b) up or down; namely, we consider the maps $(a, b) \mapsto (a+1, b+1)$ and $(a, b) \mapsto (a-1, b-1)$. If we consider how $P(x)$ acts under these maps, we get the maps:

$$P(x) = (x+a)(x+b) \mapsto (x+a+1)(x+b+1) = ((x+1)+a)((x+1)+b) = P(x+1)$$

$$P(x) = (x+a)(x+b) \mapsto (x+a-1)(x+b-1) = ((x-1)+a)((x-1)+b) = P(x-1)$$

Thus, under the maps, P is simply translated up or down by 1 unit. With this in mind, we can prove the following lemma:

Lemma. The aforementioned maps produce new valid pairs.

Where by "valid" it is meant that they verify the condition for P . The proof of the lemma is straightforward enough: for the first map, every input x gets sent to the input $x+1$; but this is still in the domain of positive integers, which because (a, b) is a valid pair satisfies the required condition, so $(a+1, b+1)$ is a valid pair. For the second map, the argument is more or less the same, with the exception that $x=1$, as that falls outside of the set of positive integers. We must thus consider the case $x=1 \mapsto x=0$; in particular, we need:

$$M_2(P(0)) = 1$$

But notice $P(0) = (0+a)(0+b) = ab$, which we know is balanced.

With our new lemma, we now set without loss of generality $b > a$ (otherwise, simply flip the pair). Write thus $b = a + c$ for a positive integer c . Using the second map a times, send the pair $(a, a+c) \mapsto (0, c)$. In this new context, consider $P(c)$:

$$P(c) = (c+0)(c+c) = 2c^2$$

By our lemma, this number must be balanced; but that is clearly a contradiction, as it is a perfect square and a prime number. Thus, c *cannot* be part of the set of positive integers; consequently, $b = a$, and we are done.

□

Side note. For the proof that ab is balanced, we are assuming both are nonzero; but given we only get *up to* 0 in our maps, it doesn't affect the proof itself.

And, for the sake of completeness, here is the "intended" solution.

Now suppose $P(n)$ is balanced for all positive integers n and $a < b$. Set $n = k(b - a) - a$ for sufficiently large k , such that n is positive. Then $P(n) = k(k + 1)(b - a)^2$, and this number can only be balanced, if $f(k) = f(k + 1)$ holds. Thus, the sequence $f(k)$ must become constant for sufficiently large k . But this is not possible, as for every prime p we have $f(p) = 1$ and for every square t^2 we have $f(t^2) = 0$. Hence $a = b$.

□

Side note. I mean... much shorter than mine. Still, I believe mine uses a "simpler" argument and isn't based on a somewhat "bright idea" by instead purely exploring and transforming a and b directly.

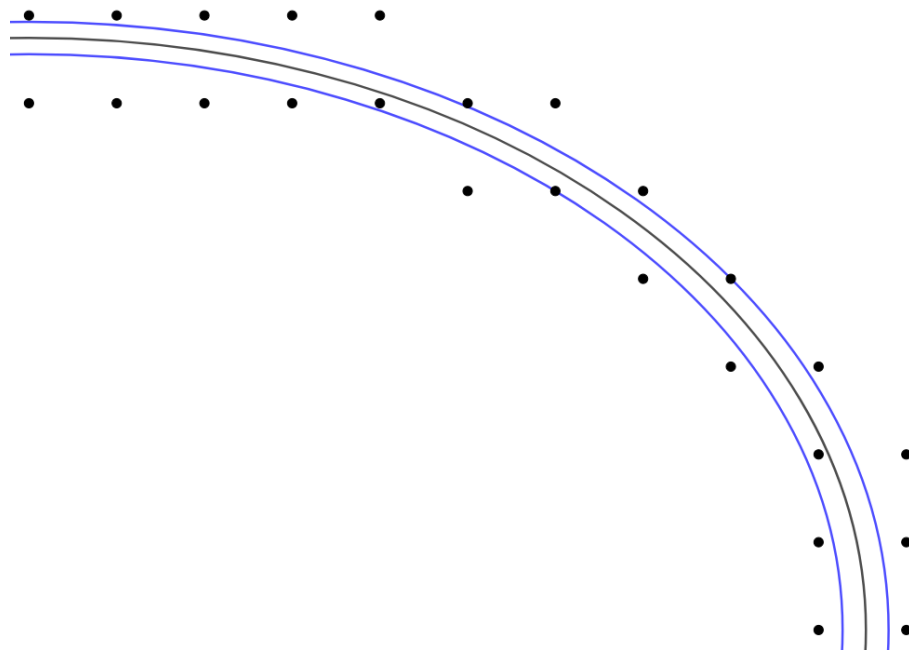
5 Putnam 1986 B-4

Problem. For a positive real number r , let $G(r)$ be the minimum value of $|r - \sqrt{m^2 + 2n^2}|$ for all integers m and n . Prove or disprove the assertion that $\lim_{r \rightarrow \infty} G(r)$ exists and equals 0.

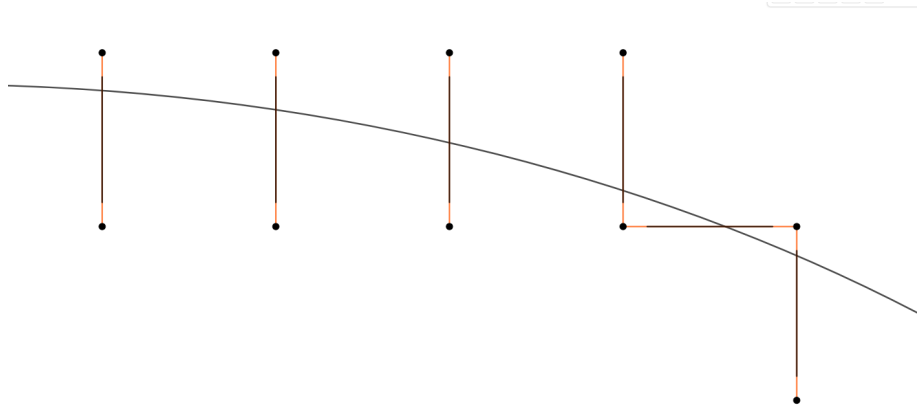
Proof. We know that the limit is well defined when given $\epsilon > 0$ we get some R such that for $r > R$, $|G(r) - L| < \epsilon$ (essentially, we get arbitrarily close). So what we want is to prove $G(r)$ *in general* gets smaller for larger r .

Now, one would want to characterise the possible values of $\sqrt{m^2 + 2n^2}$ over the integers, perhaps entering the realm of number theory. However, it seems much more productive (and somewhat surprising) to actually make use of geometry for this limit (sorry mew, geometry is just that good).

First, we consider the general conic $x^2 + 2y^2 = r^2$ over $\mathbb{R}^2$. This is clearly an ellipse, and what we'll want is to find close lattice points (recall that the lattice is the set of points with integer coordinates). In a sense, what we want to say is that as r gets larger this equates to simply scaling up the ellipse; and that as the ellipse gets larger, the lattice points get in some way *denser* around it, and that this means they get closer to it. What we will prove is that if we imagine in some way thickening the ellipse by ϵ on each side then some point will fall in this thicker line for $r > R$ (which is the definition we've stated for the limit).



The first observation we can make is that if the ellipse in fact wasn't closer than ϵ it would always have to intersect every square it passed within the middle $1 - 2\epsilon$ segment of each side (the black segments) as otherwise it'd have a point closer than ϵ .



In this context, we can see that when it goes down a square the first time it will also have passed another horizontally previously (as we can see in the figure, the last three segments); it is this passing through a corner behaviour that we will exploit.

It is clear that between the intersection horizontally and the intersection vertically there is an average slope of at least $\frac{\epsilon}{1} = \epsilon$ (in fact clearly a bit more, but this is enough). Thus, if we have an average slope of ϵ , we need a point of slope at least ϵ . What we prove is that we can ensure the second intersection has slope less than ϵ , thereby making the average slope strictly less than ϵ (as the slope before the second intersection is smaller, by the shape of the ellipse itself), and ensuring at least one lattice point is closer than ϵ .

To do so, we simply observe that the first instance of the ellipse intersecting a horizontal line must fall within the y values of $\frac{r}{\sqrt{2}}$ and $\frac{r}{\sqrt{2}} - 1$. Of these, the one with largest slope is the latter; thus, consider the point on the ellipse with y value $\frac{r}{\sqrt{2}} - 1$. Solving for x (we'll take x to be negative so the slope is positive, but we could change that without affecting the proof):

$$x^2 + 2 \left(\frac{r}{\sqrt{2}} - 1 \right)^2 = r^2 \Rightarrow x = -\sqrt{r^2 - r^2 + 2\sqrt{2}r - 2} = -\sqrt{2\sqrt{2}r - 2}$$

Now, considering the derivative of this value, we find:

$$\frac{d}{dx} \frac{\sqrt{r^2 - x^2}}{\sqrt{2}} = \frac{-2x}{2\sqrt{2}\sqrt{r^2 - x^2}} = \frac{-2 \left(-\sqrt{2\sqrt{2}r - 2} \right)}{2\sqrt{2}\sqrt{r^2 - \left(-\sqrt{2\sqrt{2}r - 2} \right)^2}} =$$

$$= \frac{\sqrt{2\sqrt{2}r - 2}}{\sqrt{2}\sqrt{r^2 - 2\sqrt{2}r + 2}}$$

Which does look like a mess, but can be resolved simply by observing the top has degree $\frac{1}{2}$ and the bottom has degree 1. Thus, this is a bottom-heavy fraction, and tends to 0 at infinity. Thus, by definition, for each ϵ there is an R such that for all $r > R$ this fraction takes a value strictly less than ϵ . In particular, this means there is an R such that the first intersection of the ellipse with a horizontal line has a slope less than ϵ , and so the ellipse must get closer than ϵ to the ellipse. Given this is an upper bound for the value of $G(r)$, and we also know $0 \leq G(r)$, the Squeeze Theorem kicks in by taking the limit; giving a limit for $G(r)$, and proving $\lim_{r \rightarrow \infty} G(r) = 0$.

□

Side note. It is clear that with almost no changes to the proof we can achieve a similar result for any $|r^2 - \sqrt{am^2 + bn^2}|$, which is a pleasant generalisation. The solution the Berkeley University gives involves only inequalities, but I enjoy this approach because it felt as if every observation was well motivated, instead of the way they prove it, which simply throws an inequality at you which works. Also, the geometric idea feels more elegant and visually pleasing. To prove my point (no pun intended), I also include the proof they give.

Proof by Berkeley. The assertion is true. Let M be a fixed positive integer and $M \leq r < M + 1$; we prove $0 \leq G(r) \leq \frac{3}{\sqrt{M}}$, which clearly goes to 0 as $M \rightarrow \infty$. Specifically, we prove that $\sqrt{M^2 + 2(n+1)^2} - \sqrt{M^2 + 2n^2} < \frac{3}{\sqrt{M}}$ whenever n is an integer with $M \leq \sqrt{M^2 + 2n^2} < M + 1$, so that every r is within $\frac{3}{\sqrt{M}}$ of some $\sqrt{M^2 + 2n^2}$. To do this, we first solve the condition $\sqrt{M^2 + 2n^2} < M + 1$ for n , yielding $0 \leq n < \sqrt{M+1/2}$, i.e. $0 \leq n \leq \sqrt{M}$. Then rationalizing the numerator yields

$$\begin{aligned} \sqrt{M^2 + 2(n+1)^2} - \sqrt{M^2 + 2n^2} &= \frac{4n+2}{\sqrt{M^2 + 2(n+1)^2} + \sqrt{M^2 + 2n^2}} < \frac{2n+1}{M} \leq \\ &\leq \frac{2}{\sqrt{M}} + \frac{1}{M} \leq \frac{3}{\sqrt{M}} \end{aligned}$$

□

6 Rapidfire Putnam 1987 problems

Problem. Let F be the field of p^2 elements, where p is an odd prime. Suppose S is a set of $\frac{p^2-1}{2}$ distinct nonzero elements of F with the property that for each

$a \neq 0$ in F , exactly one of a and $-a$ is in S . Let N be the number of elements in the intersection $S \cap \{2a : a \in S\}$. Prove that N is even.

Proof. It is well-known that the field F is isomorphic to the finite field $\mathbb{Z}/p^2\mathbb{Z}$. Consider for each nonzero $a \in F$ the map $S \mapsto (S - \{a\}) \cup \{-a\}$. After a straightforward casework it is determined quickly that this map keeps the parity of N . Thus, by building an even case we prove all cases are even.

Consider then $S = \{1, 2, \dots, \frac{p^2-1}{2}\}$. Considered as normal integers, it is clear that exactly $\frac{p^2-1}{4}$ of these elements are even, whilst the rest are odd. It is also clear that each even element has its precursor (its division by 2) within S , but that no odd element has its precursor in S (by size). Thus, there are exactly $\frac{p^2-1}{4}$ pairs $(a, 2a)$ in S . But given the quadratic residues modulo 8 are 0, 1 and 4, the result of the aforementioned division is even, finishing the proof.

□

Problem. Let $(x_1, y_1) = (0.8, 0.6)$ and let $x_{n+1} = x_n \cos y_n - y_n \sin y_n$ and $y_{n+1} = x_n \sin y_n + y_n \cos y_n$ for $n = 1, 2, 3, \dots$. For each of $\lim_{n \rightarrow \infty} x_n$ and $\lim_{n \rightarrow \infty} y_n$, prove that the limit exists and find it or prove that the limit does not exist.

Proof. If one is particularly observant, the initial values $(0.8, 0.6)$ and the recursion formula should be a big indicator of the direction of the proof. It is clear that the initial values come from the well-known Pythagorean triple 3-4-5, and that $0.8^2 + 0.6^2 = 1$. Thus, there is some angle α_1 such that $\cos \alpha_1 = x_1$ and $\sin \alpha_1 = y_1$. From here, we will prove by induction that for each pair (x_n, y_n) there is a corresponding angle α_n such that $\cos \alpha_n = x_n$ and $\sin \alpha_n = y_n$. The base case has been covered, and assuming the k -th case we now proceed to the $k+1$ -th case.

$$\begin{aligned} x_{k+1} &= (\cos \alpha_k) \cos y_k - (\sin \alpha_k) \sin y_k = \cos(\alpha_k + y_k) \\ y_{k+1} &= (\cos \alpha_k) \sin y_k + (\sin \alpha_k) \cos y_k = \sin(\alpha_k + y_k) \end{aligned}$$

Thus we can build a recursion

$$\alpha_{k+1} = \alpha_k + y_k = \alpha_k + \sin \alpha_k$$

We aren't sure if it converges yet, but if it were to converge, we would have the relation

$$L = \lim_{n \rightarrow \infty} \alpha_n = \lim_{n \rightarrow \infty} (\alpha_{n-1} + \sin \alpha_{n-1}) = L + \sin L$$

Which would clearly give $L = 0, \pi$ (up to an integer multiple of 2π). As our α_1 is strictly positive, we'd intuitively have that the limit would be π .

Now for the trickier part, proving convergence. First, we evaluate the function $f(x) = x + \sin x$. The derivative of this function is clearly $1 + \cos x$, which must be at least 0; but is only 0 at 0 or π (again, up to integer multiples of 2π). Thus, the function is *strictly increasing*. This is the first ingredient for convergence. The second comes from proving the inequality:

$$\pi - \alpha > \sin \alpha$$

For $\alpha \in (0, \pi)$. The proof of this fact involves the observation that the left side spits out an angle with equal sin to α , giving the quite interesting:

$$\pi - \alpha > \sin \alpha \Rightarrow \sin(\pi - \alpha) > \sin(\sin \alpha) \Rightarrow \sin \alpha > \sin(\sin \alpha)$$

But the rightmost inequality is actually equivalent to the well-known limit of $\frac{\sin x}{x}$. Thus, since $\sin \alpha \in (0, 1]$, and in this interval the fraction is less than 1 (just compare the graphs of $\sin x$ and x in this interval), the inequality is proven.

What we obtain from this inequality is that the distance from some α up to π is never quite covered; thus, the recursion never quite reaches π . But this implies a bound on the value of α_k , which combined with the fact it is an increasing function would give it converges; and the aforementioned limit computation gives the limit

$$\lim_{n \rightarrow \infty} \alpha_n = \pi$$

(Up to a multiple of 2π , but this doesn't affect the final result). Which in turn means

$$\begin{aligned} \lim_{n \rightarrow \infty} x_n &= \lim_{n \rightarrow \infty} \cos \alpha_n = -1 \\ \lim_{n \rightarrow \infty} y_n &= \lim_{n \rightarrow \infty} \sin \alpha_n = 0 \end{aligned}$$

□

Problem. Let r , s and t be nonnegative integers and $r + s \leq t$. Prove that

$$\frac{\binom{s}{0}}{\binom{t}{r}} + \frac{\binom{s}{1}}{\binom{t}{r+1}} + \cdots + \frac{\binom{s}{s}}{\binom{t}{r+s}} = \frac{t+1}{(t+1-s)\binom{t-s}{r}}$$

Proof. Denote by $S_{s,r}^t$ the sum in question. Consider the sum $S_{s+1,r}^t$:

$$\begin{aligned} S_{s+1,r}^t &= \frac{\binom{s+1}{0}}{\binom{t}{r}} + \frac{\binom{s+1}{1}}{\binom{t}{r+1}} + \cdots + \frac{\binom{s+1}{s+1}}{\binom{t}{r+s+1}} = \\ &= \frac{\binom{s}{0}}{\binom{t}{r}} + \frac{\binom{s}{0} + \binom{s}{1}}{\binom{t}{r+1}} + \cdots + \frac{\binom{s}{s}}{\binom{t}{r+s+1}} = \\ &= S_{s,r}^t + S_{s,r+1}^t \end{aligned}$$

And we catch the Pascal relation. Using it, we can build any pair (r, s) from other pairs. In particular, we can first build the base case:

$$S_{0,r}^t = \frac{\binom{0}{0}}{\binom{t}{r}} = \frac{1}{\binom{t}{r}} = \frac{t+1}{(t+1-0)\binom{t-0}{r}}$$

Now, by induction, we assume the formula is verified for some $s = k$ (and *any* corresponding r and t). We prove the formula works for $s = k + 1$ and all its corresponding r and t . We fix some values r, t and find:

$$\begin{aligned} S_{k+1,r}^t &= S_{k,r+1}^t + S_{k,r}^t = \frac{t+1}{(t+1-k)\binom{t-k}{r+1}} + \frac{t+1}{(t+1-k)\binom{t-k}{r}} = \\ &= \frac{t+1}{t+1-k} \left[\frac{1}{\frac{t-k-r}{r+1}\binom{t-k}{r}} + \frac{1}{\binom{t-k}{r}} \right] = \frac{t+1}{(t+1-k)\binom{t-k}{r}} \left(\frac{r+1+t-k-r}{t-k-r} \right) = \\ &= \frac{t+1}{(t-k-r)\binom{t-k}{r}} = \frac{t+1}{(t-k-r)\frac{t-k}{t-k-r}\binom{t-k-1}{r}} = \frac{t+1}{(t+1-(k+1))\binom{t-(k+1)}{r}} \end{aligned}$$

Which concludes the proof. □

7 Putnam 1988 A-5

Problem. Prove that there exists a *unique* function f from the set $\mathbb{R}^+$ of positive real numbers to $\mathbb{R}^+$ such that

$$f(f(x)) = 6x - f(x)$$

Proof. Intuitively, we'd expect $f(x)$ to be a simple function, say a linear transformation (i. e. of the form $f(x) = ax + b$). If we try out a couple simple examples, the function $f(x) = 2x$ ends up being a valid solution. Now, we need to prove that this is a *unique* solution.

First, what we will do is extend the relation given to arbitrary compositions of the functions. So let $f^n(x)$ be the function f composed n times with itself. In particular:

$$f^n = \underbrace{f \circ f \circ \cdots \circ f}_n$$

Then it is clear that from the relation given we get the generalised version:

$$f^{n+2}(x) = 6f^n(x) - f^{n+1}(x)$$

Now, we fix a value d . We consider now the *generating function* of the sequence $(f^{n+2}(d))_{n \geq 0}$. That is, we consider the function:

$$A(x) = \sum_{n=0}^{\infty} f^{n+2}(d)x^n$$

Using our generalised recursive relation:

$$\begin{aligned} A(x) &= 6x^2 A(x) - xA(x) + 6xf(d) + 6d - f(d) \\ (1 + x - 6x^2)A(x) &= 6xf(d) - f(d) + 6d \\ A(x) &= \frac{6xf(d) - f(d) + 6d}{1 + x - 6x^2} \end{aligned}$$

We can factor the bottom and use partial fractions:

$$A(x) = \frac{6xf(d) - f(d) + 6d}{(1 + 3x)(1 - 2x)}$$

$$A(x) = -\frac{9}{5} \frac{f(d) - 2d}{1 + 3x} + \frac{4}{5} \frac{f(d) + 3d}{1 - 2x}$$

But notice that both of these terms are geometric sequence limits! Thus, using the standard expansion for geometric sequences, we can group terms of equal degree of x to obtain the relation

$$f^{n+2}(d) = -\frac{9}{5}(f(d) - 2d)(-3)^n + \frac{4}{5}(f(d) + 3d)2^n$$

Notice now the first term in the right hand side. If f is $2x$ as we expect, we'd obtain that it vanishes at every value of d . Thus it is natural to attempt to prove that the first term must be zero. To do so, we'll leverage the fact that $f^{n+2}(d) > 0$ by definition. In particular, let $c_1 = f(d) - 2d$ and $c_2 = f(d) + 3d$. By definition, $c_2 > 0$, but we must break up the rest into cases:

- $c_1 > 0$ We use the even powers of $(-3)^n$, so the sign of the first term is negative. Thus, write $n = 2k$. Consider now the limit:

$$\lim_{k \rightarrow \infty} \frac{\frac{9}{5}(c_1)(-3)^{2k}}{\frac{4}{5}(c_2)2^{2k}} = \frac{9c_1}{4c_2} \lim_{k \rightarrow \infty} \frac{9^k}{4^k}$$

Which clearly grows unbounded, meaning there is a value k for which $-\frac{9}{5}c_1(-3)^{2k} + \frac{4}{5}c_22^{2k} < 0$, which is a contradiction.

- $c_1 < 0$ We now use the odd powers of $(-3)^n$, in a similar fashion. Writing $n = 2k - 1$ and considering the same limit:

$$\lim_{k \rightarrow \infty} \frac{\frac{9}{5}(c_1)(-3)^{2k-1}}{\frac{4}{5}(c_2)2^{2k-1}} = \frac{9c_1}{4c_2} \lim_{k \rightarrow \infty} \frac{-3^{2k-1}}{2^{2k-1}}$$

Which is again unbounded, and gives the same contradiction of size.

Thus, we must have $c_1 = 0$, as then the first term vanishes. Thus, $f(d) - 2d = 0$, and we obtain the desired $f(x) = 2x$.

□

8 Putnam 1989 B-3

Problem. Let f be a function on $[0, \infty)$, differentiable and satisfying

$$f'(x) = -3f(x) + 6f(2x)$$

for $x > 0$. Assume that $|f(x)| \leq e^{-\sqrt{x}}$ for $x \geq 0$ (so that $f(x)$ tends rapidly to 0 as x increases). For n a nonnegative integer, define

$$\mu_n = \int_0^\infty x^n f(x) dx$$

(sometimes called the n th moment of f).

a) Express μ_n in terms of μ_0 .

b) Prove that the sequence $\{\mu_n \frac{3^n}{n!}\}$ always converges, and that the limit is 0 only if $\mu_0 = 0$.

Proof. For the first part, we consider the following:

$$\mu_n = \int_0^\infty x^n f(x) dx$$

With the change of variable $u = \frac{x}{2}$:

$$\int_0^\infty (2u)^n f(2u) 2du = 2^{n+1} \int_0^\infty u^n f(2u) du$$

Since u is a dummy variable, we can observe, using linearity of the integral:

$$\begin{aligned} -3\mu_n + \frac{3}{2^n}\mu_n &= -3 \int_0^\infty x^n f(x) dx + 6 \int_0^\infty x^n f(2x) dx = \\ &= \int_0^\infty x^n f'(x) dx \end{aligned}$$

Using integration by parts, with $u = x^n$ and $dv = f'(x)dx$:

$$\begin{aligned} \int_0^\infty x^n f'(x) dx &= [x^n f(x)]_0^\infty - \int_0^\infty nx^{n-1} f(x) dx = \\ &= -n\mu_{n-1} \end{aligned}$$

Since the first term evaluated at 0 is 0 and evaluated at ∞ , we prove by induction it vanishes to 0 too. Since f grows at most like $e^{-\sqrt{x}}$, we prove that the limit of

$$\lim_{x \rightarrow \infty} x^n e^{-\sqrt{x}}$$

Is 0. First, for $n = 0$, this is a trivial case; then, assuming the case $n - 1$, we find the case n :

$$\begin{aligned} \lim_{x \rightarrow \infty} \frac{x^n}{e^{\sqrt{x}}} &= \lim_{x \rightarrow \infty} \frac{nx^{n-1}}{\frac{e^{\sqrt{x}}}{2\sqrt{x}}} = \lim_{x \rightarrow \infty} \frac{2n(n - \frac{1}{2})x^{n-\frac{3}{2}}}{\frac{e^{\sqrt{x}}}{2\sqrt{x}}} = \lim_{x \rightarrow \infty} \frac{4n(n - \frac{1}{2})x^{n-1}}{e^{\sqrt{x}}} \\ &= 4n(n - \frac{1}{2}) \lim_{x \rightarrow \infty} \frac{x^{n-1}}{e^{\sqrt{x}}} \end{aligned}$$

Using a double application of L'Hôpital's rule. Since the case $n - 1$ is 0, the case n is too, and we have our induction.

Thus, we have the recursion

$$\mu_n = \frac{2^n n}{3 \cdot 2^n - 3} \mu_{n-1}$$

Which gives a product:

$$\mu_n = \mu_0 \prod_{k=1}^n \frac{2^k k}{3(2^k - 1)} = \frac{\mu_0 n!}{3^n} \prod_{k=1}^n \frac{2^k}{2^k - 1}$$

Giving μ_n in terms of μ_0 .

For the second part, we have to consider the limit of:

$$\lim_{N \rightarrow \infty} \mu_0 \prod_{k=1}^N \frac{2^k}{2^k - 1}$$

Clearly the constant term μ_0 can be taken out of the limit, which leaves only the product to work with. Since products are awful to work with, we can convert this to a sum applying the natural logarithm:

$$\ln \left(\lim_{N \rightarrow \infty} \prod_{k=1}^N \frac{2^k}{2^k - 1} \right) = \lim_{N \rightarrow \infty} \ln \left(\prod_{k=1}^N \frac{2^k}{2^k - 1} \right) = \lim_{N \rightarrow \infty} \sum_{k=1}^N \ln \left(\frac{2^k}{2^k - 1} \right)$$

Notice we must prove this sum converges, not to what it converges. Thus, we may apply the ratio test to gaze if it converges without bothering to find the

exact limit. Thus, we find (ignoring the absolute values, since each term is positive anyway):

$$\lim_{n \rightarrow \infty} \frac{\ln \left(\frac{2^{n+1}}{2^{n+1}-1} \right)}{\ln \left(\frac{2^n}{2^n-1} \right)}$$

Notice the top and bottom both tend to 0 (the inside of the natural logarithms tends to 1) so we may apply L'Hôpital's rule.

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{\ln 2 - \frac{\ln 2 \cdot 2^{n+1}}{2^{n+1}-1}}{\ln 2 - \frac{\ln 2 \cdot 2^n}{2^n-1}} \\ \lim_{n \rightarrow \infty} \frac{1 - \frac{2^{n+1}}{2^{n+1}-1}}{1 - \frac{2^n}{2^n-1}} \end{aligned}$$

Notice this again satisfies the conditions to apply L'Hôpital's rule

$$\lim_{n \rightarrow \infty} \frac{\frac{\ln 2 \cdot 2^{n+1}(2^{n+1}-1) - \ln 2 \cdot 2^{n+1} 2^{n+1}}{(2^{n+1}-1)^2}}{\frac{\ln 2 \cdot 2^n(2^n-1) - \ln 2 \cdot 2^n 2^n}{(2^n-1)^2}}$$

Which looks horrendous, but we can simplify a bit

$$\lim_{n \rightarrow \infty} \frac{\frac{2^{n+1}}{(2^{n+1}-1)^2}}{\frac{2^n}{(2^n-1)^2}}$$

Reordering and simplifying even more, we get

$$2 \lim_{n \rightarrow \infty} \frac{(2^n - 1)^2}{(2^{n+1} - 1)^2}$$

Now we multiply by 1 twice:

$$2 \lim_{n \rightarrow \infty} \frac{(2^n - 1)^2}{(2^{n+1} - 1)^2} \cdot 1 \cdot 1$$

But notice 1 can be rewritten as the limit of $\frac{2^n}{2^n-1}$ when n goes to infinity, and so it can be written as the square of this limit too:

$$2 \lim_{n \rightarrow \infty} \frac{(2^n - 1)^2}{(2^{n+1} - 1)^2} \cdot \lim_{n \rightarrow \infty} \left(\frac{2^n}{2^n - 1} \right)^2 \cdot \lim_{n \rightarrow \infty} \left(\frac{2^{n+1} - 1}{2^{n+1}} \right)^2$$

But combining all of these limits:

$$2 \lim_{n \rightarrow \infty} \left(\frac{2^n}{2^{n+1}} \right)^2$$

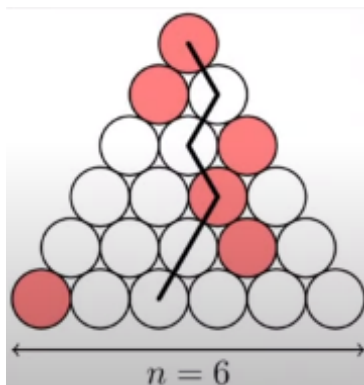
But notice this simply approaches $\frac{1}{2}$. Since this is strictly less than 1, we know the sequence converges absolutely, which means that the product also does, converging to some constant L . It is important to note this is a nonzero constant, since the elements at each step are strictly larger than 1; thus the product is never smaller than 2. Thus, $\mu_0 L$ is nonzero unless $\mu_0 = 0$.

□

Side note. It is pretty clear that this problem, at least the first part, was built with a very specific method of solving in mind, but it was nonetheless an interesting problem that made me struggle a lot with the limit (admittedly, because I was overcomplicating myself).

9 IMO 2023 P5

Problem. Let n be a positive integer. A Japanese triangle consists of $1 + 2 + \dots + n$ circles arranged in an equilateral triangular shape such that for each $i = 1, 2, \dots, n$, the i^{th} row contains exactly i circles, exactly one of which is coloured red. A ninja path in a Japanese triangle is a sequence of n circles obtained by starting in the top row, then repeatedly going from a circle to one of the two circles immediately below it and finishing in the bottom row. Here is an example of a Japanese triangle with $n = 6$, along with a ninja path in that triangle containing two red circles. In terms of n , find the greatest k such that in each Japanese triangle there is a ninja path containing at least k red circles.



Proof. We prove it is exactly $\lfloor \log_2 n \rfloor + 1$.

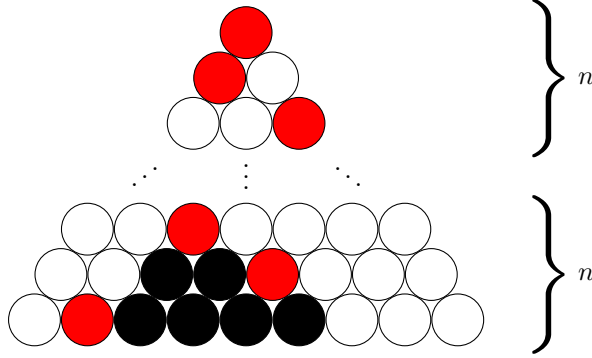
The proof is by induction. Let k_n be the k for each n . It is clear that in a Japanese triangle of 1 circle ($n = 1$) we have $k_1 = 1$ since there is only one circle and it is red. We also must check the case $n = 2$, which is a simple casework of the two possibilities, both giving the bound $k_2 = 2$. From there, we build a construction proving that $k_{2n} = k_n + 1$. Since $k_{n+1} \geq k_n$ (i. e. the sequence is nondecreasing) the statement $k_{2n} = k_n + 1$ is sufficient for the proof.

Label by T_n a Japanese triangle of height n , and by $T_n^{\rightarrow|}$ the circles of the last row of T_n where we can finish with a ninja path of maximal red circles. We label by $S_{T_n}^r$ the total number of red and black circles in row r of the Japanese triangle T_n . Also, we call a ninja path *maximal* if it contains the maximum amount of red circles possible in T_n , and we call a Japanese triangle *minimal* if its maximal ninja path(s) contain k_n red circles.

By strong induction, assume for all $m < 2n - 1$ we have $k_m = \lfloor \log_2 m \rfloor + 1$. Consider a Japanese triangle of height $2n - 1$. We break it down into two cases.

(i) If $k_{n-1} = k_n$, we show that $k_{2n-2} = k_{2n-1} = k_{2n}$, and since the sequence is nondecreasing, this is doable simply by constructing a minimal triangle.

We build it as follows. Construct a minimal n row triangle first. We know that the maximum amount of red circles in a ninja path is going to be k_n . Now, we expand the triangle by the following procedure: in the $n + 1$ -th row, place a red circle anywhere, and from this row onward we paint the circles black if they are immediately below a red circle or immediately below a black circle. We add red circles in each subsequent row in a white circle adjacent to a black circle (as in the figure) up to the $2n - 2$ -th row. It is a routine task to verify that using this construction we see that $S_{T_{n+1}}^{n+1} = 1$ and that at each row step we add 2; i. e. $S_{T_{n+1+r}}^{n+1+r} = 1 + 2r$. We find thus that constructing this triangle is always possible, since $S_{T_{n+1+r}}^{n+1+r} < n + 1 + r$ when $r < n$, which is the case for T_{2n-2} . At T_{2n} , in fact, it is still possible, since $2n = n + 1 + (n - 1)$. Thus, there is always space to place a red circle and so we have constructed a Japanese triangle with $2n$ rows and, by construction, a ninja path of no more than $\lfloor \log_2 2n \rfloor + 1$, as well as a subtriangle of size $2n - 1$ that also has this bound. Thus, the inductive step is done.



(ii) We now treat the case $k_{n-1} + 1 = k_n$. We again make use of a size argument with $S_{T_n}^r$. We first show that a triangle of size $2n$ must have more than k_n red circles in some ninja path, and then we construct a triangle that has exactly $k_n + 1$ red circles at most in one path.

The size argument is as follows. Consider any triangle T_{2n} , and consider its subtriangle T_n , with its corresponding $T_n^{\rightarrow 1}$. If within T_n we have some ninja path that contains more than k_n red circles, the bound is already verified; so suppose otherwise (that is, suppose T_n is minimal). When we consider now the subtriangle T_{2n-2} , we see that between rows $n + 1$ and $2n - 2$ we *must* have collected no red circles (otherwise, again, the bound is met). If we paint black circles from row n onward (not including the n -th row) but painting a circle black if: (1) it's immediately below one of the circles of $T_n^{\rightarrow 1}$, (2) it's immediately below a red circle that lies in the rows $n + 1$ to $2n - 2$ inclusive or (3) it's immediately below a black circle, we see that the minimum value for $S_{T_{2n}}^{2n-2}$ is $1 + 2n - 4 = 2n - 3$; which makes it possible to place a red circle in row $2n - 1$ but not in row $2n$ by size. Thus, since the *minimum* value for $S_{T_{2n}}^{2n}$ is too large, any other value will be.

The construction of the minimal triangle has in fact already been realised: we know that a minimal triangle of size $2n - 1$ is possible (shown just above) so adding one more row adds at most and exactly one red circle to a ninja path. Thus, the inductive step is done.

With both cases covered, the proof is done.

□

Side note. While the actual proof is very elementary, the details and the method used are so hard to put into words. Apparently, this problem made a lot of people from my country's IMO team lose points, since it's quite a lucky idea problem.

10 Putnam 1990 A-4

Problem. Consider a paper punch that can be centered at any point of the plane and that, when operated, removes from the plane precisely those points whose distance from the center is irrational. How many punches are needed to remove every point?

Proof. Clearly we require at least 2 punches, since 1 is insufficient. Once you punch centered at some point O , suppose you punch again at some point O_1 . Set up a coordinate system centered at O with $\overline{OO_1}$ as x-axis. Then $O = (0, 0)$ and $O_1 = (d, 0)$. After the first punch at O , only the points (x, y) such that $\sqrt{x^2 + y^2} \in \mathbb{Q}$. After the second punch, we prove some points remain.

Assume otherwise. We prove $\exists x, y$ such that both $\sqrt{x^2 + y^2}$ and $\sqrt{(x - d)^2 + y^2}$ are rational.

Select $b, c \in \mathbb{Q}$ such that

$$c^2 > b^2 > \frac{d^2}{4} + 1, c^2 - b^2 < 2d$$

These squares must exist, since the rational squares are dense over the real numbers. Set $a = c^2 - b^2$, $x = \frac{d}{2} - \frac{a}{2d}$ and $y = \sqrt{b^2 - x^2}$. Notice y can exist since $x^2 < \frac{d^2}{4} + 1 < b^2$ by construction. By construction, we'll get both square roots to be rational.

Thus, we've proven we require at least 3 punches. Such a procedure is provided.

We punch at some point $O = (0, 0)$ and then punch at $O_1 = (1, 0)$. Since after the first punch all points remaining verify $\sqrt{x^2 + y^2} \in \mathbb{Q}$. If $x \notin \mathbb{Q}$, then $\sqrt{(x - 1)^2 + y^2} = \sqrt{(x^2 + y^2) + 1 - 2x}$ cannot be rational since the interior of the square root is two rational numbers and an irrational number.

Thus only points with rational x remain. Punch a third time at $O_2 = (\pi, 0)$. Since all remaining points have $x \in \mathbb{Q}$, we have that if $\sqrt{(x - \pi)^2 + y^2} \in \mathbb{Q}$ then

$$\begin{aligned} \sqrt{\pi^2 - 2x\pi + (x^2 + y^2)} &= a \Rightarrow \\ \pi^2 - 2x \cdot \pi + [(x^2 + y^2) - a^2] &= 0 \end{aligned}$$

is a rational polynomial with π as a root, which is impossible since π is transcendental. Thus, all points have been punched out.

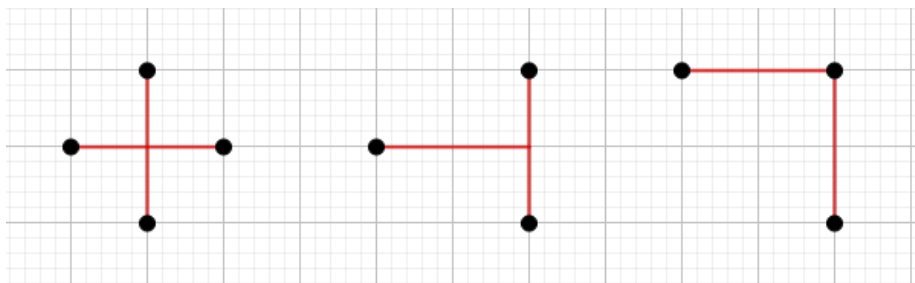
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Side note. The statement of the problem made me say "there's no way this is solvable" at first. Nonetheless, it is not only solvable, but relatively approachable. Overall, quite an interesting problem to solve.

11 Putnam 1990 A-3

Problem. Prove that any convex pentagon whose vertices (no three of which are collinear) have integer coordinates must have area greater than or equal to $5/2$.

Proof. We show a pentagon as described contains a subpolygon of area $5/2$. First, notice that since we have 5 points no three collinear, we must have that the longest chord horizontally and the longest chord vertically have length at least 2 each. Since it is a convex polygon, the chords must meet somewhere. Thus, there are two chords of length 2 perpendicularly intersecting each other. This leaves 3 possible setups:



In all of these, we see there are at most 4 total vertices in the hull of these diagonals. Thus, there is at least 1 more vertex of the pentagon outside of the hull. Since any triangle made by vertices on integer coordinates can be readily verified to have area no less than $1/2$ (say, by Heron's formula), the vertex outside the hull will form a triangle that will add $1/2$ to these setups. Since all of these setups have area 2, the total is thus of at least $5/2$.

□

12 Putnam 1991 B-2

Problem. Suppose f and g are non-constant, differentiable, real valued functions defined on $(-\infty, \infty)$. Furthermore, suppose that for each pair of real numbers x and y ,

$$f(x+y) = f(x)f(y) - g(x)g(y)$$

$$g(x+y) = f(x)g(y) + g(x)f(y)$$

If $f'(0) = 0$, prove that $(f(x))^2 + (g(x))^2 = 1$ for all x .

Proof. First, consider a case $x = y$, and take the derivative of both given equations (which we can do because all elements involved are differentiable):

$$\begin{aligned} D[f(2x)] &= D[f^2(x) - g^2(x)] \Rightarrow 2f'(x) = 2f'(x)f(x) - 2g(x)g'(x) \\ D[g(2x)] &= D[2f(x)g(x)] \Rightarrow 2g'(x) = 2f'(x)g(x) + 2f(x)g'(x) \end{aligned}$$

From the first one, we can see that at $x = 0$ we get

$$2f'(0) = 2f(0)f'(0) - 2g(0)g'(0) \Rightarrow -2g(0)g'(0) = 0$$

Thus at least one of $g(0), g'(0)$ is 0.

Further, from the second at $x = 0$ we get

$$2g'(0) = 2f'(0)g(0) + 2g'(0)f(0) \Rightarrow g'(0) = g'(0)f(0)$$

Thus at least one of $g'(0) = 0, f(0) = 1$ is true.

Now, consider the double partial derivative of the first expression:

$$\frac{\partial^2}{\partial x \partial y} f(x+y) = \frac{\partial^2}{\partial x \partial y} [f(x)f(y) - g(x)g(y)] \Rightarrow f''(x+y) = f'(x)f'(y) - g'(x)g'(y)$$

Which at $y = 0$ gives

$$f''(x) = g'(x)g'(0)$$

If the latter was 0 (as suggested by the previous conditions) then $f''(x) = 0$ for all x , meaning $f'(x)$ is constant. Since $f'(0) = 0$, $f'(x) = 0$ for all x , which contradicts the nonconstant condition.

Thus, $g(0) = 0$ and $f(0) = 1$.

Label by $\alpha = g'(0)$ the value at 0 of the derivative of g .

Consider now the partial derivative wrt y of the two expressions:

$$\frac{\partial}{\partial y} g(x+y) = \frac{\partial}{\partial y} [f(x)g(y) + g(x)f(y)] \Rightarrow g'(x+y) = f(x)g'(y) + g(x)f'(y)$$

Which at $y = 0$ gives

$$g'(x) = \alpha f(x)$$

Similarly, the partial wrt y of the other expression gives

$$f'(x) = -\alpha g(x)$$

To finish, consider the derivative of:

$$D[(f(x))^2 + (g(x))^2] = 2f(x)f'(x) + 2g(x)g'(x)$$

Substituting:

$$2\alpha(-f(x)g(x) + f(x)g(x)) = 0$$

Thus the sum of squares is constant. Given we know the values at $x = 0$, we find:

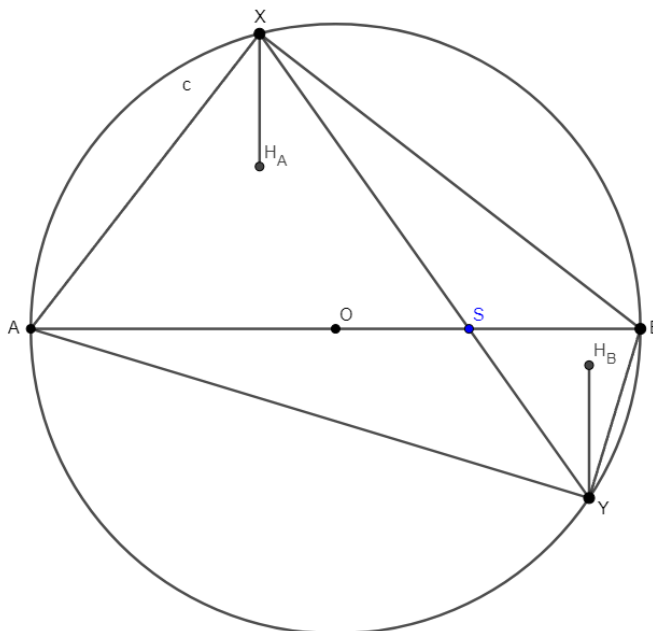
$$(f(0))^2 + (g(0))^2 = 1^2 + 0^2 = 1$$

Finishing the proof.

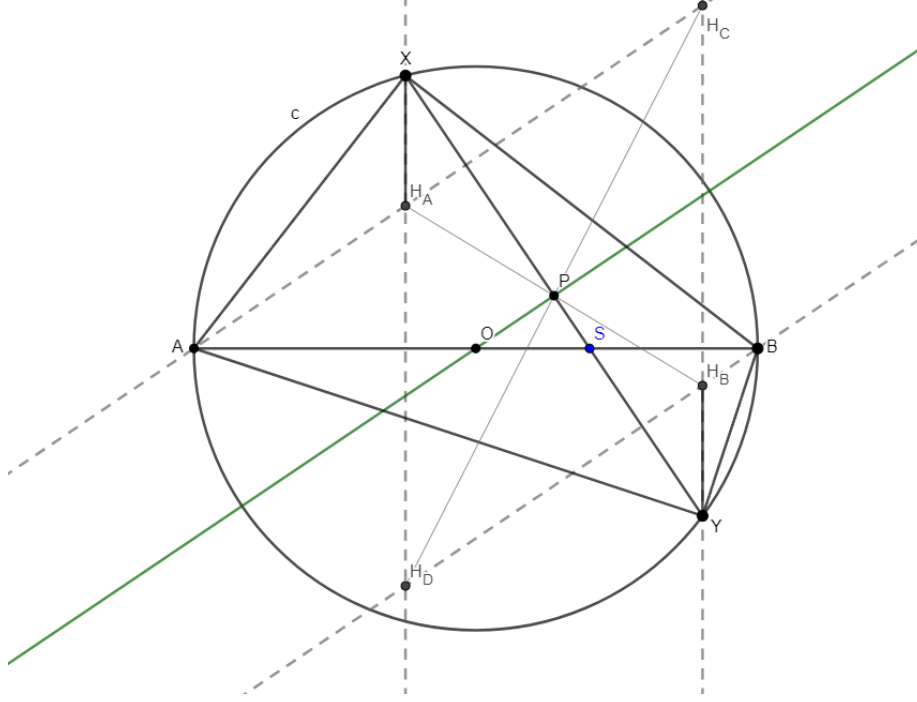
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13 OMG 2022 3

Problem. Let $AXBY$ be a quadrilateral such that $\angle AXB = \angle BYA = 90^\circ$. Let S be the intersection of AB and XY . Let H_A, H_B be the orthocenters of triangles $\triangle XAS, \triangle YBS$ respectively. Show that $XHA = YHB$.



Proof. Let H_C, H_D be the orthocenters of ASY, XSB . By construction, $H_A H_B H_C H_D$ is a parallelogram. Let O be the center of the circle $(AXBY)$ (which exists by the angle condition) and let P be the intersection of $H_A H_B$ with $H_C H_D$.



Claim. The line OP is parallel to the lines $H_A H_D$ and $H_B H_C$.

Proof Consider the reflection over O . Then $A \mapsto B$, and thus $H_A H_D \mapsto H_B H_C$. Since the reflection over P also sends $H_A \mapsto H_B$ and $H_C \mapsto H_D$, it also equidists from $H_A H_D$ and $H_B H_C$. Thus, OP is parallel to both of these lines.

Thus, given $\triangle XOY$ is isosceles and $OP \perp XY$, X and Y equidist from OP . When reflecting over P , X goes to a point a distance of $\frac{XY}{2}$ from the line OP (since this stays constant) and on the line $H_B H_D$, but this point is exactly Y . Thus, since reflection over P sends $XH_A \mapsto YH_B$, these must have the same length.

□

14 Putnam 1994 A-5

Problem. Let $(r_n)_{n \geq 0}$ be a sequence of positive real numbers such that $\lim_{n \rightarrow \infty} r_n = 0$. Let S be the set of numbers representable as a sum

$$r_{i_1} + r_{i_2} + \cdots + r_{i_{1994}}$$

with $i_1 < i_2 < \cdots < i_{1994}$. Show that every nonempty interval (a, b) contains a nonempty subinterval (c, d) that does not intersect S .

Proof. Note this is in fact the standard definition of a dense set over an open interval (an open set in the euclidean topology of $\mathbb{R}$). We in fact prove that no set of all sums of some finite length N is dense over an open interval.

We do so by an "anti-inductive" argument of sorts: we prove that for the result to hold we require induction, but that there is a base case that is *false*, so all cases are false.

First, we know that no finite set can be dense over an open interval (this can be proven in a multitude of ways but is in itself quite obvious). We consider now some number $r \in \mathbb{R}^+$ and some $\varepsilon > 0$, and consider the interval $(r - \varepsilon, r)$. For a sum of our form of N terms to lie in this interval, we need at least one of the terms of the sum to verify $r_{i_{k_1}} \geq \frac{r - \varepsilon}{N}$. Since $\lim_{n \rightarrow \infty} r_n = 0$, we know that there are finitely many terms in (r_n) that verify this bound, unless the bound is less than 0. Fixing some $r_{i_{k_1}} \geq \frac{r - \varepsilon}{N}$, we also have some $r_{i_{k_2}} \geq \frac{r - \varepsilon - r_{i_{k_1}}}{N - 1}$, which there are finitely many of unless the bound is less than 0, and so on. We conclude that unless there is at least 1 sum that arrives to a bound less than 0 in less than N terms. This means that this sum verifies that it lies in the interval $(r - \varepsilon, r)$ and uses less than N terms, which we can convert to exactly $N - 1$ terms using the fact that $\lim_{n \rightarrow \infty} r_n = 0$. Thus, if sums of length N lie in every open interval, then sums of length $N - 1$ lie in every open interval.

Since the case $N = 1$ breaks down (for *any* bound larger than 0 we'll have finitely many elements, and any open interval either contains a positive or negative element so we have *some* bound), we have that no set of sums of a given finite length work.

□

15 Putnam 1995 B-1

Problem. For a partition π of $\{1, 2, 3, 4, 5, 6, 7, 8, 9\}$, let $\pi(x)$ be the number of elements in the part containing x . Prove that for any two partitions π and π' , there are two distinct numbers x and y in $\{1, 2, 3, 4, 5, 6, 7, 8, 9\}$ such that $\pi(x) = \pi(y)$ and $\pi'(x) = \pi'(y)$. [A partition of a set S is a collection of disjoint subsets (parts) whose union is S .]

Proof. We proceed by considering the equivalence classes of the partitions π and π' defined by the relations $x \sim y \iff \pi(x) = \pi(y)$ and $x \sim' y \iff \pi'(x) = \pi'(y)$. These are trivially equivalence relations, and so they form corresponding equivalence classes. Denote by n_π the class such that $x \in n_\pi \iff \pi(x) = n$. It is clear that the maximal amount of simultaneous classes for a given set $\{1, 2, \dots, n\}$ is

$$f(n) := \max \left(m : \frac{m(m+1)}{2} \leq n \right)$$

(i. e. the largest m such that $1 + 2 + \dots + m \leq n$). For 9, this value is 3 ($1 + 2 + 3 + 4 > 9$). Thus, if there is one class that contains more than 3 elements, two elements of this class must end up in the same class on the other partition (by pigeonhole). We show this is necessary.

Clearly this is true if we have less than 3 classes since the total amount of elements is 9, so at least one class would have at least 4 elements. If there are exactly 3 classes, we observe each class contains exactly 3 elements. Thus, by size, these classes must be $1_\pi, 2_\pi, 3_\pi$. But since the class 2_π must contain an even number of elements, we are done.

□

Side note. As the equivalence classes method seemed relatively simple to generalise, I tried to play around with it to see what else I could find. However, it seems that 9 is very special in this regard, since as the numbers grow we have more and more freedom to set up the partitions and they look reasonably doable (it casually coincides that $\frac{n}{m} = m$ for 9, but this is far from the case for larger numbers). In essence, we find that if there is a partition such that we avoid the coinciding pair, then there is a way to break up n into a sum of the form

$$1x_1 + 2x_2 + \dots + mx_m = n$$

And such that $x_k \leq \lfloor \frac{n}{k} \rfloor$. I, however, have no idea how one would prove if such a sum is possible to construct for a general n . A simple sub case is when $\frac{m(m+1)}{2} = n$, since we can make each $x_k = 1$ and be easily done (for example, we can easily set up such a partition with 6: one set is (1), (2, 3) and (4, 5, 6) and the other set is (4), (5, 2), (6, 3, 1), in which no pair verifies the condition). However, for more generic objects, we're entering a moderate level of combinatorics knowledge, and combinatorics is probably my weakest point in mathematics. So, for now, it stays in a drawer until I know how to tackle this.

16 Rapidfire IMO 1995 SL

Problem. G2. ABC is a triangle. Show that there is a unique point P such that

$$PA^2 + PB^2 + AB^2 = PB^2 + PC^2 + BC^2 = PC^2 + PA^2 + CA^2$$

Proof. The proof is by a simple construction argument. Consider the triangle PAB . By the cosine law, we know $PB^2 = PA^2 + AB^2 - 2\cos(\angle PAB)$. Thus, $\cos(\angle PAB) = \frac{PB^2 - PA^2 - AB^2}{2}$. If we had the condition given by the problem, we can see that $(PB^2 + PC^2 + BC^2) - (PC^2 + PA^2 + CA^2) = 0$, so $PB^2 - PA^2 = CA^2 - BC^2$. This would mean $\cos(\angle PAB) = \frac{CA^2 - BC^2 - AB^2}{2}$.

Thus, we also get $\sin(\angle PAB) = \sqrt{1 - \left(\frac{CA^2 - BC^2 - AB^2}{2}\right)^2}$. This gives us that $\sin(\angle PAB) = \sin(\angle PCB)$ with the same kind of calculations; and similarly we get $\sin(\angle PBA) = \sin(\angle PCA)$ and $\sin(\angle PAC) = \sin(\angle PBC)$. These equalities translate easily into the trigonometric version of Ceva's Theorem, proving that if we constructed such a point P it would be unique. Since we've found angles independent from P , we can simply find the point P that verifies such angles by reversing the procedure described.

□

Problem. N1. k is a positive integer. Show that there are infinitely many squares of the form $2^k n - 7$.

Proof. This one took me an embarrassingly long time to find the pattern, when the proof is actually a two liner. The proof goes: We prove there is a square congruent to -7 modulo 2^k by induction. Label by x_k such a square for 2^k . The base case is $k = 1$, for which $1^2 \equiv -7$. Then, we find two possibilities:

- (i) If $x_k^2 \equiv -7 \pmod{2^{k+1}}$, we already have a valid $x_{k+1} = x_k$.
- (ii) Else we have $2^k n_k - 7 = x_k^2$ meaning $x_k^2 \equiv 2^k - 7 \pmod{2^{k+1}}$. Consider $(2^{k-1} - x_k)^2 \equiv 2^{2k-2} - 2^k x_k + x_k^2 \equiv x_k^2 - 2^k x_k$. Since x_k is odd by definition, we have $x_k^2 - 2^k x_k \equiv 2^k - 7 - 2^k \equiv -7$. Thus, a valid square exists.

□

Problem. G3. ABC is a triangle. The incircle touches BC, CA, AB at D, E, F respectively. X is a point inside the triangle such that the incircle of XBC touches BC at D . It touches CX at Y and XB at Z . Show that $EFZY$ is cyclic.

Proof. I must admit that I did use GeoGebra to get an intuition for what to do, but for the rest the proof is quite simple outside of understanding geometric inversion.

The idea of the proof is as following: set a circle centered at D with radius r . Then considering inversion over this circle (say Ω), we get that $E^*F^*||Y^*Z^*$

because their circles pass through D and are tangent to the same line. What one observes in GeoGebra is that the lines EY and FZ are also parallel and perpendicular to BC (and constant over all positions of X !). We then simply prove that in fact the distance from Y to the perpendicular to BC at D is constant.

We prove it with the properties of inverses and by trig (ouch). By inverses, the height at which the line YZ is found easily by $\frac{r^2}{2r(DYZ)}$ (where $r(ABC)$ denotes the radius of circle (ABC)). The angle $\angle I_X DY$ is equal to $\angle I_X CD$ by cyclic quads (Where I_X is the center of the incircle of XBC). We can find the $\sin \angle XBC$ easily with $\frac{r(DYZ)}{\sqrt{CD^2 + r(DYZ)^2}}$ by definition and Pythagoras. We can see clearly as well that then $\tan \angle XBC = \frac{\sin(\angle XBC)}{\sqrt{1 - (\sin \angle XBC)^2}}$, so $\tan \angle XBC = \frac{\frac{r(DYZ)}{\sqrt{CD^2 + r(DYZ)^2}}}{\sqrt{1 - (\frac{r(DYZ)}{\sqrt{CD^2 + r(DYZ)^2}})^2}} = \frac{r(DYZ)}{CD}$. Finally, we simply find that the distance is $\frac{r^2}{2r(DYZ)} \tan \angle XBC = \frac{r^2}{2CD}$, which is constant. Thus, it must be that E^*, Y^* lie in a line perpendicular to BC , and similarly with F^*, Z^* . Thus, these four points lie in a rectangle, which is cyclic by definition; after undoing the inversion, we'll get another circle $(EYZF)$.

□

17 IMO 1995 SL G8

Problem. G8. ABC is a triangle. A circle through B and C meets the side AB again at C' and meets the side AC again at B' . Let H be the orthocenter of ABC and H' the orthocenter of $AB'C'$. Show that the lines BB' , CC' and HH' are concurrent.

Proof. This one is very very meh. Using GeoGebra, one gets some very key intuitions; namely, the degree of the curve the concurrency point was on. Before that, it was simple to see that by the definition of the triangle $AB'C'$, it is similar to ABC (by angles on a cyclic quad). Thus, it is easy to see as well that H' lies always in the line defined as the reflection of AH over the A -bisector. Now, with GeoGebra, one can begin observing that the concurrency point (say P) seems to move on a hyperbola. With moving points, it is not difficult to see this is the case: moving H' linearly, we build the point B' by the perpendicular to AB intersecting AC . A small sine law and Thales' theorem shows B' also moves linearly. Connecting the lines HH' and BB' we get two degree 1 lines (they're a constant and a linearly moving point), which thus intersect at a point of degree at most 2. It's easily checked to be nonconstant and nonlinear (take the cases $B' = A, B' = C, BB' \perp AC$, all very trivial). Thus, it moves on a conic. Which one? We find five points: $B' = A \rightarrow P = A$, $B' = C \rightarrow P = C$, $BB' \perp AC \rightarrow P = H$, $H' \in (ABC)$, which gives $P = H'$ because the line AH' in fact is the isogonal of AH and goes through O . Thus we get the five points of

the conic completely independently from H' ; this is clearly symmetrically true for C' . We label the conic by $\mathcal{C}$. We now simply check that B

18 IMO 1997 SL 14

Problem. 14. $b > 1$ and $m > n$. Show that if $b^m - 1$ and $b^n - 1$ have the same prime divisors then $b + 1$ is a power of 2. [For example, $7 - 1 = 2 \cdot 3$, $7^2 - 1 = 2^4 \cdot 3$]

Proof. The proof is involved, but quite elementary in nature. First, we convert n into $\text{lcm}(\text{ord}_{p_i}(b) \mid p_i \mid b^n - 1)$. Since all orders are kept, we lose no prime factors; since this number is less than n , we don't add prime factors. Since it is, in some sense, the order of b modulo all of these primes, we get $n \mid m$. We get that we can write $m = nk$ for $k \in \mathbb{Z}^+$. Suppose k is composite. Write $k = qk_0$, where $q \in \mathbb{P}$. Then $b^n - 1 \mid b^{nq} - 1 \mid b^m - 1$, so $b^n - 1$ shares all prime factors with $b^{nq} - 1$. Thus, the problem reduces to $b^n - 1$ and $b^{nq} - 1$.

Now notice that $\frac{b^{nq}-1}{b^n-1}$ consists of at least one prime power. Since we know the factorisation of the top and bottom, it is easy to see with some simplification that it equals $b^{(q-1)n} + b^{(q-2)n} + \dots + b^n + 1$. Consider some prime p that divides this sum. Since it is part of the factors of $b^{nq} - 1$, it must be part of the factors of $b^n - 1$, so $b^n \equiv 1 \pmod{p}$. Thus, considering the sum modulo p , we get $\underbrace{1 + 1 + \dots + 1 + 1}_q \equiv 0 \pmod{p}$. This means $p \mid q$, but since both are prime

numbers, we get $p = q$, and this is the only prime number dividing this sum.

Now, since $b > 1$, we have $\frac{b^{nq}-1}{b^n-1} > q$, so $q^2 \mid b^{(q-1)n} + b^{(q-2)n} + \dots + b^n + 1$. Since $b^n \equiv 1 \pmod{q}$, we get $b^n = xq + 1$ for some $x \in \mathbb{Z}^+$. Considering now the sum modulo q^2 , expanding binomials, deleting higher degrees and grouping terms, we get:

$$(xp + 1)^{q-1} + (xp + 1)^{q-2} + \dots + (xp + 1)^1 + 1 \equiv xp \frac{(q-1)q}{2} + q \equiv 0 \pmod{q}$$

Which gives $q \equiv 0 \pmod{q}$, a contradiction, unless $q = 2$. Thus we get $b^n - 1$ and $b^{2n} - 1$. Consider now $x = v_2(n)$ and compare $b^{2^x} - 1$ and $b^{2^{x+1}} - 1$. If $p \mid b^{2^{x+1}} - 1$ but $p \nmid b^{2^x} - 1$, then $p \mid b^{2^n} - 1$ but $p \nmid b^n - 1$, which contradicts assumption. Thus, we reduce to the case $b^{2^x} - 1, b^{2^{x+1}} - 1$.

From here, it is straightforward: since $2 \mid b^{2^x} - 1$, b is odd. Since $b^{2^x} + 1 = 2^k$ (it is the only prime number that divides it) and $k > 1$ (by the previous argument) we get $2 \parallel b^{2^x} - 1$ (where $\parallel$ indicates exact division). But if $x > 0$, we get $b^{2^x} - 1 = (b^{2^{x-1}} - 1)(b^{2^{x-1}} + 1)$, with both factors multiples of 2; this is a contradiction, so $x = 0$. Thus, we get $b - 1, b^2 - 1$, which gives $\frac{b^2-1}{b-1} = b+1 = 2^k$, and we are done.

□

Side note. There are a couple of rephrasings and changes that lead to the same result. Firstly, we could write this proof as "trivial using Zsigmondy's Theorem" and leave in peace, since this is quite literally a sub case of that theorem. The other option is to skip one step by citing a very unexpected theorem. When we have $b^n - 1$ and $b^{2^n} - 1$, we can skip directly to the conclusion by seeing that $b^n + 1 = 2^k$ is a case of Catalan's Theorem. Using it, we see $n = 1$ directly, since the only possibility would be $3^2 - 2^3 = 1$, which isn't applicable. From there, we skip the entire reduction using $v_2(n)$, but if we cite Catalan we could have cited Zsigmondy by that point. Still, it is interesting to see that two seemingly unrelated theorems are useful in the same problem.

19 IMO 1997 SL 15

Problem. If an infinite arithmetic progression of positive integers contains a square and a cube, show that it must contain a sixth power.

Proof We have that if d is the difference in the arithmetic sequence and a is the initial term, then $a \equiv x^2 \equiv y^3 \pmod{d}$ for some x, y . We prove this implies $\exists z: a \equiv z^6 \pmod{d}$.

By CRT, it is clear that if we prove the condition for prime powers we prove it for any number. So consider $a \equiv x^2 \equiv y^3 \pmod{p^k}$. Make x, y be between 0 and $p^k - 1$, and we can write them uniquely as $x_0 p^{m_x}, y_0 p^{m_y}$ where x_0, y_0 are coprime with p . Then:

$$x_0^2 p^{2m_x} = y_0^3 p^{3m_y} + n p^k$$

Since the RHS is exactly divisible by p^{3m_y} , it is clear $p^{2m_x} = p^{3m_y}$, so we can write the exponent as p^{6m} . Furthermore, by size, we can write $0 < a, b < p^{k-6m}$ such that $a \equiv x_0^2 \pmod{p^k}$, $b \equiv y_0^3 \pmod{p^k}$. Thus:

$$(a - b)p^{6m} = n p^k$$

But since $|a - b| < p^{k-6m}$, we get $a = b$ and $n = 0$. Thus, we get $x_0^2 \equiv y_0^3 \pmod{p^k}$. We can then write $g^{n_x} \equiv x_0 \pmod{p^k}$, $g^{n_y} \equiv y_0 \pmod{p^k}$, $0 \leq n_x, n_y < (p-1)p^{k-1}$, g being the primitive root of p^k , which we know exists.

Now, since $x^2 \equiv a \iff (-x)^2 \equiv a \pmod{p^k}$ and $g^{\frac{(p-1)p^{k-1}}{2}} \equiv -1 \pmod{p^k}$, and $3n_y < 3(p-1)p^{k-1}$, we can guarantee that $|2n_x - 3n_y| < (p-1)p^{k-1}$, with an appropriate change $x_0 \rightarrow -x_0$ if necessary. Once we get $|2n_x - 3n_y| < (p-1)p^{k-1}$, since $|2n_x - 3n_y| = r(p-1)p^{k-1}$ by definition, we get $2n_x = 3n_y$, which means we can write the exponent as g^{6t} . Finally, we conclude that $a \equiv (g^t p^m)^6 \pmod{p^k}$, so a is a sixth power residue, and by CRT we get the conclusion that a is a sixth power residue modulo d , and we are done.

□

20 USAMO 2003 P5

Problem. Let a, b, c be positive real numbers. Prove that

$$\frac{(2a+b+c)^2}{2a^2+(b+c)^2} + \frac{(2b+c+a)^2}{2b^2+(c+a)^2} + \frac{(2c+a+b)^2}{2c^2+(a+b)^2} \leq 8$$

Proof. WLOG, we may assume $a \geq b \geq c$. Consider:

$$f(a, b, c) = \frac{(2a+b+c)^2}{2a^2+(b+c)^2} + \frac{(2b+c+a)^2}{2b^2+(c+a)^2} + \frac{(2c+a+b)^2}{2c^2+(a+b)^2}$$

We may now compute:

$$\begin{aligned} \frac{\partial}{\partial a} f(a, b, c) &= \frac{4(2a+b+c)(2a^2+(b+c)^2) - (2a+b+c)^2 4a}{(2a^2+(b+c)^2)^2} \\ &\quad + \frac{2(2b+c+a)(2b^2+(c+a)^2) - (2b+c+a)^2 2(c+a)}{(2b^2+(c+a)^2)^2} \\ &\quad + \frac{2(2c+a+b)(2c^2+(a+b)^2) - (2c+a+b)^2 2(a+b)}{(2c^2+(a+b)^2)^2} \end{aligned}$$

This looks like a monster, but we really only care about the numerators. Considering only them, with some work, we see that they are all negative when:

$$a > \max\left(\frac{b+c}{2}, b-c, c-b\right)$$

But this is true always unless $a = b = c$, since if either $a > b \geq c$ or $a \geq b > c$ we must verify the inequality. Thus, consider the case $a = b = c$:

$$f(a, a, a) = 3 \frac{(4a)^2}{2a^2 + (2a)^2} = 8$$

So we have equality. Since any other case must have $a > \max(\frac{b+c}{2}, b-c, c-b)$, the partial derivative shows us the function peaks at 8 and only goes lower. Thus $f(a, b, c) \leq 8$, and we are done.

□

Side note. Indeed, I am not dead! Finally did something new for the math forum, and hopefully I'll get some time to do some more in the future. For now, enjoy this seemingly short proof that involves university calculus techniques and an expansion that looks horrifying but actually collapses to something very neat.

21 Putnam 2023 A2

Problem. Let n be an even positive integer. Let p be a monic, real polynomial of degree $2n$; that is to say, $p(x) = x^{2n} + a_{2n-1}x^{2n-1} + \cdots + a_1x + a_0$ for some real coefficients $a_0, \dots, a_{2n-1}$. Suppose that $p\left(\frac{1}{k}\right) = k^2$ for all integers k such that $1 \leq |k| \leq n$. Find all other real numbers x for which $p\left(\frac{1}{x}\right) = x^2$.

Proof. Given $q(x) = p(x) - p(-x) = 2a_{2n-1}x^{2n-1} + 2a_{2n-3}x^{2n-3} + \cdots + 2a_1x$. And since for all $1 \leq |k| \leq n$ we have $p\left(\frac{1}{k}\right) = p\left(-\frac{1}{k}\right) = k^2$, we get that such k are roots of the polynomial $q(x)$. But $q(x)$ is a degree $2n-1$ polynomial, and we have $2n$ roots for it; thus, it is in fact the 0 polynomial. Therefore, for all odd i , $a_i = 0$.

Thus $p(x)$ is an even function. To solve for $p\left(\frac{1}{x}\right) = x^2$ is to solve for $p(x) = \frac{1}{x^2}$, or equivalently $h(x) = x^2p(x) - 1 = 0$. By the rules of even functions, this last function is even as well, and also a polynomial of degree $2n+2$ and monic. Since we have $2n$ of its roots already, we can only have 2 other roots, with them being of the form $\alpha, -\alpha$. Thus, by the Vieta formulae, we get that $(1)(-1)(2)(-2) \cdots (n)(-n)(\alpha)(-\alpha) = -1$. This lets us obtain $\alpha = \frac{1}{\sqrt{\prod_{k=1}^n k^2}}$, where the sign of the product inside the radical is positive given the parity of n .

□

22 Putnam 2023 A4

Problem. Let $v_1, \dots, v_{12}$ be unit vectors in $\mathbb{R}^3$ from the origin to the vertices of a regular icosahedron. Show that for every vector $v \in \mathbb{R}^3$ and every $\varepsilon > 0$, there exist integers $a_1, \dots, a_{12}$ such that $\|a_1v_1 + \cdots + a_{12}v_{12} - v\| < \varepsilon$.

Proof. The proof proceeds by constructing three linearly independent vectors of arbitrarily small size, and showing they can be used to approximate arbitrarily well any vector in $\mathbb{R}^3$.

Consider a vertex of the icosahedron, which has 5 adjacent triangles. Since these are equilateral triangles all of same side length, the triangles form a regular pentagon through their bases. It is well known that the ratio of a regular pentagon's diagonal to its side is $\varphi = \frac{1+\sqrt{5}}{2}$, the golden ratio. Since we have the vectors to any vertex of the icosahedron, we may take the difference between any two of them to obtain the side across them, so we can obtain both the side of the pentagon and its diagonal such that these two vectors are collinear (call them $\vec{s}, \varphi\vec{s}$). We prove that $\|a\vec{s} + b\varphi\vec{s}\|$ for $a, b \in \mathbb{Z}$ can get arbitrarily close to 0.

Lemma. $\|a\vec{s} + b\varphi\vec{s}\|$ for $a, b \in \mathbb{Z}$ can get arbitrarily close to 0.

Proof. Consider the sequence $|F_{n+1} - F_n\varphi|$, where F_n is the n -th Fibonacci number. Then the sequence is monotone decreasing, since:

$$\begin{aligned}
|F_{n+1} - F_n\varphi| < |F_n - F_{n-1}\varphi| &\iff (F_{n+1} - F_n\varphi)^2 - (F_n - F_{n-1}\varphi)^2 < 0 \iff \\
& \text{(after some simplification)} \\
&\iff 2F_n^2 + 2F_nF_{n-1} - 2F_n^2\varphi - F_{n-1}^2\varphi < 0 \iff \\
&\iff (2 - 2\varphi) \left(\frac{F_n}{F_{n-1}} \right)^2 + 2 \left(\frac{F_n}{F_{n-1}} \right) - \varphi < 0
\end{aligned}$$

This last inequality is essentially a quadratic $(2 - 2\varphi)x^2 + 2x - \varphi$, which has discriminant:

$$\Delta = 4 + 4\varphi(2 - 2\varphi) = 4 + 8\varphi - 8\varphi^2 = -4 < 0$$

Thus the quadratic has no roots, meaning the sign is always the same (otherwise, we'd have a root by Bolzano's Theorem). Therefore, taking $x = 0$, we see $-\varphi < 0$ implies the inequality.

Thus $|F_{n+1} - F_n\varphi|$ is monotone decreasing. Given it's at least 0 (since it's an absolute value) it converges. We prove this convergence is to 0. Consider:

$$\lim_{n \rightarrow \infty} |F_{n+1} - F_n\varphi| = \frac{1}{\sqrt{5}} \lim_{n \rightarrow \infty} |\varphi^{n+1} - \varphi^{-n-1} - (\varphi^n - \varphi^{-n})\varphi| = \frac{1}{\sqrt{5}} |\varphi^{-n}|$$

Which converges to 0 since $\varphi > 1$. Thus, the sequence $|F_{n+1} - F_n\varphi|$ gets arbitrarily close to 0. Thus, $\|F_{n+1}\vec{s} - F_n\varphi\vec{s}\| = |F_{n+1} - F_n\varphi| \|\vec{s}\|$ can be arbitrarily close to 0.

Suppose now we have v and $\varepsilon > 0$.

Using the lemma, we may now take e_1, e_2, e_3 three sides of the icosahedron that are linearly independent. Thus, they form a basis of $\mathbb{R}^3$, so we may write v as $\alpha_1 e_1 + \alpha_2 e_2 + \alpha_3 e_3$. Taking now N such that $\varepsilon' = |F_{N+1} - F_N\varphi| < \frac{\varepsilon}{\|e_1\| + \|e_2\| + \|e_3\|}$, we can consider

$$v' = \left\lfloor \frac{\alpha_1}{\varepsilon'} \right\rfloor \varepsilon' e_1 + \left\lfloor \frac{\alpha_2}{\varepsilon'} \right\rfloor \varepsilon' e_2 + \left\lfloor \frac{\alpha_3}{\varepsilon'} \right\rfloor \varepsilon' e_3$$

Which is an integer linear combination of our vectors. But

$$\begin{aligned}
\|v - v'\| &\leq \|(\alpha_1 - \left\lfloor \frac{\alpha_1}{\varepsilon'} \right\rfloor \varepsilon') e_1\| + \|(\alpha_2 - \left\lfloor \frac{\alpha_2}{\varepsilon'} \right\rfloor \varepsilon') e_2\| + \|(\alpha_3 - \left\lfloor \frac{\alpha_3}{\varepsilon'} \right\rfloor \varepsilon') e_3\| < \\
&< \varepsilon' (\|e_1\| + \|e_2\| + \|e_3\|) = \varepsilon
\end{aligned}$$

And the bound is met.

□

Side note. For this side note, I wanted to do a full view into how, what and where everything in this proof comes from; a little glimpse into my empty brain. This one is going to be a long one, so strap in.

Firstly, why this problem? Why not go for the functional equation in A3 or any of the other problems in the Putnam 2023? Why not even go for a different olympiad entirely?

The answer to this is quite deep, but it boils down to the type of problem and the olympiad. I chose the Putnam 2023 because it's, well, a Putnam exam, that is quite recent (which, in theory, makes it harder than the older ones) and also because it took place on my birthday, which obviously made me more interested in it. As for the problem itself, it was chosen because it is a geometry problem, which is relatively rare at this level (although it definitely goes in a topology/calculus direction instead, but those fields also interest me quite a lot, as other problems show), and because the geometric figure used, the icosahedron, is one I also used in my Extended Essay in the IB, so I felt very interested to work with it.

Now then, what lead me to the proof written above? I should first clarify that this problem took me around 3 days of work off and on (mostly off) to solve. My first approach went in the direction of trying to prove that the space is dense, i. e. in a more topological approach. Thus I first established another density condition: that I can place a vector arbitrarily close to a line through $(0, 0, 0)$. Hopefully, that would lead me to any point then.

And indeed I proved that I could get any line, using an argument inspired by a couple of sources. Firstly, I played around with just the idea of the vectors adding up together. That led me to the sum of three vertices of a face, which gives the vector for that triangle's centroid, just scaled thrice. Which gave me this idea of using the lines instead of the points. Big jump in logic, but pretty justified when considering my previous experience with barycentric coordinates and the projective plane, which is defined as taking the lines through $(0, 0, 0)$ and intersecting them with a plane (so pretty much the same idea).

Then, my motivating idea was the math IA for the IB that a friend of mine did, on what points can be generated on a triangle using certain operations. Given I could take the centroid of three points (or a vector along the line through $(0, 0, 0)$ and the centroid, but the idea stands) I could find a point inside of the triangle of these three points. So I could find any point by just taking subtriangles and their centroid and subtriangles and their centroid... until I was arbitrarily close (I like to think of this process as a binary search, as it greatly reminds me of the computer algorithm of the same name).

And so I had it! I could get arbitrarily close to any point on my triangle, in the sense that I could get any line that intersected that triangle and went through the origin. A lot of blabbery, but I was getting closer. Now what I wanted was to take an arbitrarily small vector in any of these lines. So I thought of taking

two lines that were very very close to each other, subtract the vectors that were along these lines, and I'd get a tiny one, right?

Quite a sensible thought, but of course, never that simple. It could work, but only if I had even a rough sense of scale of the size of my "very close together vectors". They had to 1) have relatively similar size and 2) have a size that wasn't so big it nullified the effects of the small angle between them. Since I had no approach to either, I wasn't going anywhere.

My brain turned back to the statement of the problem. Hell, why would they use an icosahedron for such a thing? Well it is a regular polyhedron... so symmetries? So we could go with taking a point that was close to one of the axes of symmetry of the triangles (which are equilateral, so the symmetry lies along the medians), and take advantage of the symmetry of the triangle to have another vector reflected across that axis of symmetry and whose size is equal. So we solved issue 1!

But what about issue 2? Well, since the idea had no relation to the size, I was stuck again.

The issue here was that my vectors were nowhere near aligned (that's why I tried small angle differences instead). So why not literally force that? A reflection across the median produces a point that is perpendicular to the median itself, as is obvious. So if I used the 6 symmetries of the triangle, I could have 2 pairs of points, both reflected over the median, and both forming vectors of different sizes. And that would be great, except if one vector is just an integer multiple of the other, as then I gain nothing. So I checked the ratio of the vectors in terms of some subvariables and played around with it, using some cartesian coordinates with an example point and got... an integer multiple. Not only that, but a straightforward proof that any reflection pair produced one. That was... not fun.

So back to the drawing board. By this point, my arguments have led me to some kind of overarching idea, though. Specifically, the use of two vectors of irrational ratio... which led me to the above proof. I noticed the pentagon on the icosahedron because I knew the Schläfli symbol of it is $\{3, 5\}$, which I only knew from my Extended Essay for the IB! This was not some genius conclusion, but the biggest coincidence on Earth!

Anyways, I knew the ratio of a regular pentagon's diagonal to its side is the golden ratio because it's (relatively) well-known even casually if you're interested in math. And since I knew the golden ratio is irrational, I could give it a go now, since I had collinear vectors with irrational ratios!

The second part of the proof, involving the lemma, is also a series of coincidences. I knew I wanted to get something very small, so I tried to go around with $\varphi - 1, 1 - (\varphi - 1)$, etc. And since I tried no further than 3 such examples, I of course established the pattern was $|(n + 1) - n\varphi|$. Obviously, I was wrong. I did notice this after only a bit, but it was quite dumb on my part (but eh,

it happens, right?). Anyways, I liked $|F_{n+1} - F_n\varphi|$, since it reminded me of a video I'd watched at some point that involved multiplying the golden ratio with Fibonacci numbers (which I believe was about powers of the golden ratio, but I cannot remember for certain). So it felt good.

The use of the square to avoid an absolute value is relatively normal and known in limits and approximations. The simplification and all led me to a set of products of Fibonacci numbers I had no idea how to work with. And the I remembered that the ratio of Fibonacci numbers approaches the golden ratio itself! So that was most definitely useful, and I tried to get it by dividing everything by F_{n-1}^2 . The "approaching golden ratio" idea still looked futile, but I noticed this looked like a quadratic polynomial, which even in school we are taught how to find the sign of: using the roots. I hoped for the roots to be reasonably simple... but they just didn't exist! So it was even easier than that: I could just not bother finding the intervals, and just get always negative results.

As for finding that it converges to 0, that was an afterthought. The proof isn't too crazy, though, as I did expect to need some kind of closed form formula for the Fibonacci numbers for it to be anywhere near tractable, so I did recall that there is a formula in terms of the golden ratio (really, knowing a lot about the golden ratio and the Fibonacci sequence is as surprising as nothing, given how popular this topic is in math entertainment). Then, the proof went relatively smooth sailing as far as the limit goes. The final step is pretty uninspiring. Given an arbitrarily small basis, taking an approximation with the floor function seemed as reasonable as it goes, and the inequalities are very straightforward given some thought.

So that is it. What a proof, and what a journey the thing was. Hitting walls at every step, but at least I was picking what walls I'd hit? In conclusion, one will find that the totality of the proof is based on ideas I'd seen before, or some intuition that is very reasonable within the context. So do not be mistaken: I am not brilliant in any way, or do leaps in logic worthy of a gymnastics medal. I purely take some ideas I've seen before, bang my head against every wall, until there are no walls left to hit; and get a proof.

Moving the focus toward the math again, what can we conclude? Well, I can say a couple things. For example, it is interesting to see that this problem is equivalent to the same statement when substituting "icosahedron" with "dodecahedron". This latter option would probably have led to the same proof in a much quicker way, since the faces of the dodecahedron are already pentagons. And how do I know that I can easily convert from the icosahedron to the dodecahedron? Well of course, because of my Extended Essay. The dodecahedron is the *dual* of the icosahedron, which is a fancy way to say that they're somewhat equivalent; for example, take their Schläfli symbols, $\{3, 5\}$ and $\{5, 3\}$. In fact, if you took those sums of the three vertices of a triangle for each face of the icosahedron, you'd have the 20 vertices of a dodecahedron.

Another thing to mention is that I'm pretty sure that any irrational number α verifies that $a + b\alpha$ can get arbitrarily small, probably by using some argument of coprime numbers, somewhere along the lines of $a_1 + b_1\alpha$ and $a_2 + b_2\alpha$ such that $\gcd(b_1, b_2) = 1$ means we can get something smaller than both of these numbers. In very unnecessarily fancy terms, I'm saying that I think $\mathbb{Z}[\alpha]$ is dense over $\mathbb{R}$ for any irrational α .

And that would be it! I hope this wasn't just an unnecessarily long side note but rather a detailed breakdown of how it's possible for me to find proofs for things: just look up a lot of stuff that you can later use.

23 Putnam 1999 A3

Problem. Consider the power series expansion

$$\frac{1}{1 - 2x - x^2} = \sum_{n=0}^{\infty} a_n x^n$$

Prove that, for each integer $n \geq 0$, there is an integer m such that

$$a_n^2 + a_{n+1}^2 = a_m$$

Proof. The proof is as straightforward as one would expect. We observe that the fraction may be written as

$$\frac{1}{1 - 2x - x^2} = \frac{1}{2 - (x+1)^2} = \frac{1}{(\sqrt{2} - 1 - x)(\sqrt{2} + 1 + x)}$$

Using partial fractions, this equates to

$$\frac{1}{2\sqrt{2}(\sqrt{2} - 1 - x)} + \frac{1}{2\sqrt{2}(\sqrt{2} + 1 + x)}$$

Obviously, these are equivalent to the infinite sums of a geometric series (recall that that was of the form $\frac{a_0}{1-r}$, so we're quite close to it). In particular, they equate to

$$\sum_{n=0}^{\infty} \frac{1}{2\sqrt{2}(\sqrt{2} - 1)^{n+1}} x^n - \sum_{n=0}^{\infty} \frac{1}{2\sqrt{2}(-\sqrt{2} - 1)^{n+1}} x^n$$

This leads us to the final equality

$$a_n = \frac{1}{2\sqrt{2}(\sqrt{2} - 1)^{n+1}} - \frac{1}{2\sqrt{2}(-\sqrt{2} - 1)^{n+1}}$$

Given this, the solution is as straightforward as expanding $a_n^2 + a_{n+1}^2$ and finding that it reduces to a_{2n+2} (I omit showing this expansion and reduction since it essentially is just a long string of simple computations and I'm too lazy for them).

□

Side note. While checking my proof to be right, I stumbled across a particularly nice alternative proof that just must be noted. The proof consists of considering that $\frac{1}{1-2x-x^2}$ leads to the recursion $a_{n+2} = 2a_{n+1} + a_n$, and one can compute the first two elements (i. e. a_0 and a_1) by setting $x = 0$ and by computing the derivative and setting $x = 0$. Thus one finds $a_0 = 1, a_1 = 2, a_{n+2} = 2a_{n+1} + a_n$. The nice idea is to write a matrix that represents the transformation $(a_n, a_{n+1}) \rightarrow a_{n+2}$. This leads to:

$$\begin{pmatrix} a_n & a_{n+1} \\ a_{n+1} & a_{n+2} \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} a_{n+1} & a_{n+2} \\ a_{n+2} & a_{n+3} \end{pmatrix}$$

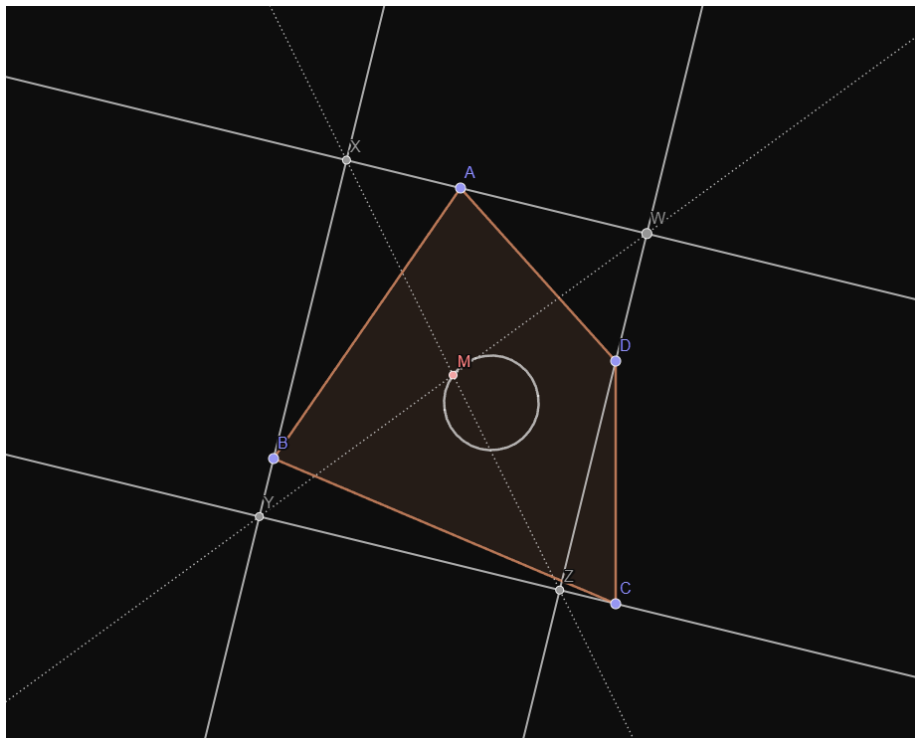
Thus the matrix $A = \begin{pmatrix} 0 & 1 \\ 1 & 2 \end{pmatrix}$ represents our transformation. But the absolute nicest part is that they find that A^2 looks like:

$$\begin{pmatrix} 0 & 1 \\ 1 & 2 \end{pmatrix}^2 = \begin{pmatrix} 1 & 2 \\ 2 & 5 \end{pmatrix} = \begin{pmatrix} a_0 & a_1 \\ a_1 & a_2 \end{pmatrix}$$

So, in fact, $A^{n+2} = \begin{pmatrix} a_n & a_{n+1} \\ a_{n+1} & a_{n+2} \end{pmatrix}$! Thus the proof concludes just by using $A^{n+2}A^{n+2} = A^{(2n+2)+2}$ and looking at the first entry of the matrix, which is equivalent to $a_n^2 + a_{n+1}^2 = a_{2n+2}$. The amount of internal patterns required for this to work just amazes me. The proof barely even manipulates anything, except perhaps one 2×2 matrix product, and is still able to find the same result without issue by abusing these patterns. Overall, a remarkable proof.

24 PERA P2

Problem. Let $ABCD$ be a quadrilateral. Let $WXYZ$ be a rectangle such that A, B, C, D lie on lines $\overline{WX}, \overline{XY}, \overline{YZ}, \overline{ZW}$, respectively. Let M be the intersection of lines $\overline{WY}$ and $\overline{XZ}$. Show that the locus of points described by M for all possible rectangles $WXYZ$ is a circle.



Proof. Through a bit of interpretation in an example drawing, one sees that the circle is the circle whose diameter are the midpoints of $\overline{AC}$ and $\overline{BD}$. To show that M lies in a circle with them, we can show that $\angle EMF$ (where E, F are the midpoints of the aforementioned lines) is constant. Using the Inscribed Angle Theorem, that leads to the circle and we'd be done.

To show $\angle EMF = 90^\circ$, we may show that $\overline{EM}$ is parallel to $\overline{WX}$. By symmetry that would imply that $\overline{MF}$ is parallel to $\overline{XY}$, so they would have the same angle, namely 90° .

The proof of this parallel abuses some interesting ideas. Consider the quadrilateral $AXCZ$. Using the Euler line, we have that the midpoints of $\overline{AC}$, $\overline{XZ}$, and of the points $\overline{AX} \cap \overline{CZ}$, $\overline{XC} \cap \overline{ZA}$ are collinear. The first two points are E and M , while this last one is a midpoint of an infinite point (since $\overline{AX}$ and

$\overline{ZC}$ are parallel) so it stays in that infinite direction. In particular, this leads to $\overline{EM}$ being parallel to $\overline{AX}$, which easily leads to the circle, and finishes the proof.

□

25 Putnam 1990 B-4

Problem. Let G be a finite group of order n generated by a and b . Prove or disprove: there is a sequence

$$g_1, g_2, g_3, \dots, g_{2n}$$

such that

- (1) every element of G occurs exactly twice, and
- (2) g_{i+1} equals $g_i a$ or $g_i b$ for $i = 1, 2, \dots, 2n$. (Interpret g_{2n+1} as g_1 .)

Proof. This problem will, in fact, not be treated with any group theory whatsoever. Instead, we create a *directed graph* (or *digraph*) D such that it has n vertices, each labeled by an element of G , and such that the edge $\overrightarrow{uv}$ exists if $v = ua$ or $v = ub$. Then the problem equivalently translates to finding an Euler cycle in this graph, that is, a cycle that passes every edge, that goes through each vertex exactly twice. The neat part is that since the in- and outdegree of each vertex is 2 (each connects to only one multiplication by a , by b , by a^{-1} and b^{-1}) we can only ever possibly pass each vertex twice, so we just require an Euler cycle. Further, since the group is generated by a and b , we know it is also a strongly connected digraph; that is, there is a directed path connecting every pair of vertices.

But this essentially proves the problem by itself. Not only is it already known that even strongly connected digraphs have directed Euler cycles in them, but it is relatively straightforward to prove:

Consider a finite digraph D such that all vertices have equal in- and outdegree, the underlying graph is simple, and such that the nonzero degree vertices are strongly connected. Then by starting at any vertex and following the arrows we must eventually form a cycle, since the digraph is finite. Thus we may remove the arcs of the cycle, which leaves another finite digraph with all vertices of equal in- and outdegree, a simple underlying graph and a strongly connected component. So we may decompose the graph into directed cycles, and these cycles must intersect at least one other cycle. Then we can traverse recursively each cycle, entering another cycle when we find it, and thus navigate the entire digraph. Finally, we may then conclude that yes, such a sequence does exist, since the digraph that represents this group relation has an Euler cycle.

□

26 USAMO 2008 P1

Problem. (*Titu Andreescu*) Prove that for each positive integer n , there are pairwise relatively prime integers $k_0, k_1, \dots, k_n$, all strictly greater than 1, such that $k_0 k_1 \cdots k_n - 1$ is the product of two consecutive integers.

Proof. We prove this in two parts: a "Euclid's infinitude of primes" proof, and a CRT use.

Suppose, FSOC, that the sequence $a_n := n(n+1) + 1$ has finitely many prime divisors. Let P be the product of all these primes. Then $a_P := P(P+1) + 1$ is not divisible by any of them, so we get a contradiction.

Since we have infinitely many primes p such that $\exists a \in \mathbb{Z}: p \mid a(a+1) + 1$, we can use CRT on these residues to form numbers with arbitrarily many prime divisors, and thus we are done.

□

27 USAMO 2009 P1

Problem. Given circles ω_1 and ω_2 intersecting at points X and Y , let ℓ_1 be a line through the center of ω_1 intersecting ω_2 at points P and Q and let ℓ_2 be a line through the center of ω_2 intersecting ω_1 at points R and S . Prove that if P, Q, R and S lie on a circle then the center of this circle lies on line XY .

Proof. $\omega_3 := (PQRS)$

$$\mathbb{P}_{\omega_2}(O_1) = \mathbb{P}_{\omega_3}(O_1)$$

$$\mathbb{P}_{\omega_1}(O_2) = \mathbb{P}_{\omega_3}(O_2)$$

$$\mathbb{P}_{\omega_1}(O_3) = d(O_1, O_2)^2 + r_3^2 - r_1^2 - r_2^2 = \mathbb{P}_{\omega_2}(O_3)$$

□

28 Olympiad gc P1

Problem. For any $n \in \mathbb{Z}^+$, find all $\{r_0, r_1, \dots, r_n\} \subset \mathbb{Z}_{\geq 0}$ such that

$$\sum_{k=0}^n n^k \mid \sum_{k=0}^n n^{r_k}$$

Proof. It's clear $(n-1) \sum_{k=0}^n n^k = n^{n+1} - 1$, so $n^{n+1} \equiv 1 \pmod{\sum_{k=0}^n n^k}$, and we only need to treat cases with $0 \leq r_k \leq n$, as all others are modularly equivalent. Since $(n+1)n^n < n \sum_{k=0}^n n^k$ for $n > 1$, all numbers we have to consider must have all digits equal in base n , which is only possible with $\underbrace{11 \dots 1}_n$

base n . Thus, we get any set congruent to $\{0, \dots, n\}$ modulo $n+1$.

□

29 Inco Daily P9

Problem. Compute

$$\sum_{k=-\infty}^0 \frac{1}{k^2 + 2k + 2}$$

Proof. By changing the sign and completing the square, we get:

$$\begin{aligned} \sum_{k=0}^{\infty} \frac{1}{k^2 - 2k + 2} &= \sum_{k=0}^{\infty} \frac{1}{(k-1)^2 + 1} = \\ &= \frac{1}{2} + \sum_{k=0}^{\infty} \frac{1}{k^2 + 1} \end{aligned}$$

Now, we restructure this for a geometric series sum:

$$\begin{aligned} \frac{1}{2} + 1 + \sum_{k=1}^{\infty} \frac{1}{k^2 + 1} &= \frac{1}{2} + 1 + \sum_{k=1}^{\infty} \left[1 - \frac{k^2}{k^2 + 1} \right] = \\ &= \frac{1}{2} + 1 + \sum_{k=1}^{\infty} \left[1 - \frac{1}{1 + \frac{1}{k^2}} \right] = \frac{1}{2} + 1 + \sum_{k=1}^{\infty} \left[1 - \frac{1}{1 - \frac{-1}{k^2}} \right] = \\ &= \frac{1}{2} + 1 + \sum_{k=1}^{\infty} \left[1 - \sum_{m=0}^{\infty} \left(\frac{-1}{k^2} \right)^m \right] = \frac{1}{2} + 1 + \sum_{k=1}^{\infty} \sum_{m=1}^{\infty} \left(\frac{-1}{k^2} \right)^m = \\ &= \frac{1}{2} + 1 + \sum_{k=1}^{\infty} \sum_{m=1}^{\infty} \frac{(-1)^m}{k^{2m}} \end{aligned}$$

Quite beautifully, one might observe that we can reverse the order of summation to obtain a pretty nice sum of zeta functions:

$$\frac{1}{2} + 1 + \sum_{m=1}^{\infty} \sum_{k=1}^{\infty} \frac{(-1)^m}{k^{2m}} = \frac{1}{2} + 1 + \sum_{m=1}^{\infty} (-1)^m \zeta(2m)$$

Now, I think it's justified enough to use a nice series sum related to positive even zeta function values:

$$\sum_{m=0}^{\infty} \zeta(2m) x^{2m} = -\frac{\pi x}{2} \cot(\pi x)$$

We can see that this matches with our sum valued at $x = i$:

$$\frac{1}{2} + 1 + \sum_{m=1}^{\infty} \zeta(2m) i^{2m} = \frac{1}{2} + 1 + \left(-\frac{\pi i}{2} \cot(\pi i) - \zeta(0) i^0 \right)$$

Now, we use the following observations:

$$\sin(x) = \frac{e^{ix} - e^{-ix}}{2i}$$

$$\cos(x) = \frac{e^{ix} + e^{-ix}}{2}$$

$$\zeta(0) = -\frac{1}{2}$$

Therefore:

$$\begin{aligned} \frac{1}{2} + 1 + \left(-\frac{\pi i}{2} \frac{\frac{e^{ii\pi} + e^{-ii\pi}}{2}}{\frac{e^{ii\pi} - e^{-ii\pi}}{2i}} + \frac{1}{2} \right) &= \\ &= 2 + \frac{\pi}{2} \frac{e^{-\pi} + e^{\pi}}{e^{-\pi} - e^{\pi}} = \\ &= 2 + \frac{\pi}{2} \frac{1 + e^{2\pi}}{1 - e^{2\pi}} \end{aligned}$$

Finishing the result. □

30 Inco Daily P14

Problem. Compute

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} e^{e^{i \tan x}} dx$$

Proof. We expand the exponential using its Taylor series:

$$\begin{aligned} e^x &= \sum_{n=0}^{\infty} \frac{x^n}{n!} \\ \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sum_{n=0}^{\infty} \frac{e^{in \tan x}}{n!} dx &= \\ \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sum_{n=0}^{\infty} \frac{\cos(n \tan x) + i \sin(n \tan x)}{n!} dx & \end{aligned}$$

Since we're integrating bounded functions over bounded intervals, we may swap the summation and integration (each term converges):

$$\sum_{n=0}^{\infty} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\cos(n \tan x) + i \sin(n \tan x)}{n!} dx$$

$$\sum_{n=0}^{\infty} \left[\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\cos(n \tan x)}{n!} dx + i \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\sin(n \tan x)}{n!} dx \right]$$

Thankfully, the sinusoidal term simply cancels out, as it's an odd function over a symmetric interval. Thus, we're left with:

$$\sum_{n=0}^{\infty} \frac{1}{n!} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos(n \tan x) dx$$

Applying a simple $u = n \tan x$ sub:

$$du = n \sec^2 x dx \implies \frac{du}{n \left(\left(\frac{u}{n} \right)^2 + 1 \right)} = dx$$

$$x = -\frac{\pi}{2} \rightarrow u = -\infty, \quad x = \frac{\pi}{2} \rightarrow u = \infty$$

$$\sum_{n=0}^{\infty} \frac{1}{n!} \int_{-\infty}^{\infty} \frac{\cos(u)}{n \left(\left(\frac{u}{n} \right)^2 + 1 \right)} du$$

Rephrasing to simplify:

$$\sum_{n=0}^{\infty} \frac{1}{n!} \int_{-\infty}^{\infty} \frac{n \cos(u)}{u^2 + n^2} du$$

We focus on the inner integral now:

$$\int \frac{n \cos(u)}{u^2 + n^2} du$$

Using partial fractions:

$$u^2 + n^2 = (u + in)(u - in)$$

$$A(u - in) + B(u + in) = 1$$

$$u = in \implies B(2in) = 1 \implies B = \frac{1}{2in} = -\frac{i}{2n}$$

$$u = -in \implies A(-2in) = 1 \implies A = \frac{1}{-2in} = \frac{i}{2n}$$

Thus

$$\int \left[\frac{i}{2} \frac{\cos u}{u + in} - \frac{i}{2} \frac{\cos u}{u - in} \right] du$$

$$\frac{i}{2} \int \frac{\cos u}{u + in} du - \frac{i}{2} \int \frac{\cos u}{u - in} du$$

We make a change of variable for each integral:

$$\begin{aligned} v_1 &= u + in, \quad v_2 = u - in \\ dv_1 &= du, \quad dv_2 = du \\ \frac{i}{2} \int \frac{\cos(v_1 - in)}{v_1} dv_1 - \frac{i}{2} \int \frac{\cos(v_2 + in)}{v_2} dv_2 \end{aligned}$$

Expanding the cosine with the angle sum formula:

$$\frac{i}{2} \int \frac{\cos(v_1) \cos(-in) - \sin(v_1) \sin(-in)}{v_1} dv_1 - \frac{i}{2} \int \frac{\cos(v_2) \cos(in) - \sin(v_2) \sin(in)}{v_2} dv_2$$

Using $\sin(i\alpha) = i \sinh(\alpha)$, $\cos(i\alpha) = i \cosh(\alpha)$:

$$\frac{i}{2} \int \frac{\cos(v_1) \cosh(n) + i \sin(v_1) \sinh(n)}{v_1} dv_1 - \frac{i}{2} \int \frac{\cos(v_2) \cosh(n) - i \sin(v_2) \sinh(n)}{v_2} dv_2$$

Which further separates into

$$\begin{aligned} \frac{i \cosh(n)}{2} \int \frac{\cos(v_1)}{v_1} dv_1 - \frac{\sinh(n)}{2} \int \frac{\sin(v_1)}{v_1} dv_1 \\ - \frac{i \cosh(n)}{2} \int \frac{\cos(v_2)}{v_2} dv_2 - \frac{\sinh(n)}{2} \int \frac{\sin(v_2)}{v_2} dv_2 \end{aligned}$$

Which, thankfully, can be simplified using the notation $\text{Si}(z) = \int \frac{\sin z}{z} dz$, $\text{Ci}(z) = \int \frac{\cos z}{z} dz$:

$$\begin{aligned} \frac{i \cosh(n)}{2} \text{Ci}(v_1) - \frac{\sinh(n)}{2} \text{Si}(v_1) - \frac{i \cosh(n)}{2} \text{Ci}(v_2) - \frac{\sinh(n)}{2} \text{Si}(v_2) \\ \frac{i \cosh(n)}{2} (\text{Ci}(v_1) - \text{Ci}(v_2)) - \frac{\sinh(n)}{2} (\text{Si}(v_1) + \text{Si}(v_2)) \\ \frac{i \cosh(n)}{2} (\text{Ci}(u + in) - \text{Ci}(u - in)) - \frac{\sinh(n)}{2} (\text{Si}(u + in) + \text{Si}(u - in)) \end{aligned}$$

Now, we reintroduce the bounds of integration:

$$\left[\frac{i \cosh(n)}{2} (\text{Ci}(u + in) - \text{Ci}(u - in)) - \frac{\sinh(n)}{2} (\text{Si}(u + in) + \text{Si}(u - in)) \right]_{-\infty}^{\infty}$$

This is where I make an observation. Up until now, everything was more or less workable without external knowledge. Unfortunately now, we introduce the values of the limits of Si and Ci at infinity using a "trust me" method, since these *are* well-known (especially $\text{Si}(\infty)$, the Dirichlet integral) but a full proof is too long to include. However, I'll likely add an addendum where I work through these values one by one at some point in the future.

- $\lim_{z \rightarrow \infty} \text{Si}(z) = \frac{\pi}{2}$
- $\lim_{z \rightarrow -\infty} \text{Si}(z) = -\frac{\pi}{2}$
- $\lim_{z \rightarrow \infty} \text{Ci}(z) = 0$
- $\lim_{z \rightarrow -\infty} \text{Si}(z) = \pm i\pi$ (depending on $\text{sgn}(\text{Im}(z))$)

In particular, then, we obtain:

$$\left[\frac{i \cosh(n)}{2} (0 - 0) - \frac{\sinh(n)}{2} \left(\frac{\pi}{2} + \frac{\pi}{2} \right) \right] \\ - \left[\frac{i \cosh(n)}{2} (i\pi - (-i\pi)) - \frac{\sinh(n)}{2} \left(-\frac{\pi}{2} + \left(-\frac{\pi}{2} \right) \right) \right]$$

Which simply ends up as:

$$\pi \cosh(n) - \pi \sinh(n) \\ \pi \frac{e^n + e^{-n}}{2} - \pi \frac{e^n - e^{-n}}{2} \\ \pi e^{-n}$$

Finally, we can reintroduce this into our sum:

$$\sum_{n=0}^{\infty} \frac{1}{n!} \pi e^{-n} \\ \pi \sum_{n=0}^{\infty} \frac{(e^{-1})^n}{n!}$$

Which again converges as a Taylor series of e^x :

$$\pi e^{e^{-1}}$$

And we are done.

□

31 RDA Disc P1

Problem. Find all numbers $p, q, r \in \mathbb{P}$ such that:

$$p^2 + 4q^2 = r^2$$

Where $\mathbb{P}$ is the set of prime numbers.

Proof. We notice that $p, 2q, r$ is a primitive Pythagorean triple unless $p, r = 2$, but that case immediately leads to a contradiction. Thus, it is well known that we may write $(p, 2q, r)$ as $(m^2 - n^2, 2mn, m^2 + n^2)$ for some $m, n \in \mathbb{Z}$, with one of m, n even.

This leads to $2 \mid q$, so $q = 2$ (as it is prime). Therefore, we get $p^2 + 16 = r^2$, or $r^2 - p^2 = 16$. Since the difference of two consecutive squares is always odd, $r \geq p + 2$. Thus:

$$(p + 2)^2 - p^2 \leq r^2 - p^2 = 16$$

$$4p + 4 \leq 16$$

$$p \leq 3$$

At $p = 2$ we've already found contradiction, so $p = 3$ is the only solution. This gives $r = 5$, and so the only solution is $(3, 2, 5)$, which gives the classic Pythagorean triple $(3, 4, 5)$.

□

32 RDA Disc P2

Problem. Let f be a nonconstant polynomial with nonnegative integer coefficients. Show

$$f(n) \mid f(f(n) + 1) \iff n = 1$$

Proof. The proof is very direct by modular arithmetic. For $\Leftarrow$, we simply use $f(x) \equiv f(y) \pmod{m}$ when $x \equiv y \pmod{m}$, with $x = f(1) + 1, y = 1, m = f(1)$.

For $\Rightarrow$, we use the same lemma getting $f(n) \mid f(1)$, but since $f \in \mathbb{Z}_{\geq 0}[x]$ it is monotone increasing, so $n \leq 1 \Rightarrow n = 1$.

□